Conformalized Heteroscedastic Variant Effect Prediction with Local Coverage Guarantees

Anonymous Authors
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Abstract

[TODO: Draft — replace after §6 results locked.] Variant effect prediction (VEP) powers clinical interpretation of genomic variants, yet production deployments lack calibrated uncertainty. We introduce **HCCP** (Heteroscedastic Class-Conditional Conformal Prediction), a post-hoc framework that pairs a gradient-boosted aggregator over pre-trained genomic features with a learned variance head and a Mondrian conformal calibration stratified by $(y \times \hat{\sigma}\text{-bin})$. On TraitGym [Benegas et al., 2025], HCCP matches SOTA AUPRC (0.900 Mendelian, 0.350 Complex) while providing finite-sample marginal, class-conditional, and $\hat{\sigma}$-bin-local coverage. We prove these guarantees under a chrom-group exchangeability assumption and establish a bin-conditional coverage bound $1/(n_{k,b} + 1)$ compatible with Barber's (2020) pointwise impossibility. Three-axis OOD evaluation (chromosome-LOO, trait-LOO, cross-dataset) shows $\hat{\sigma}$-bin coverage gap ≤ 0.04 across all regimes, including a gap of 0.002–0.004 reaching the finite-sample floor under trait-LOO. Our ablations position HCCP as a principled strengthening of DEGU's heteroscedastic distillation [Zhou et al., 2026].

1 Introduction

Clinical variant interpretation needs more than a score. Classifying a human genomic variant as pathogenic or benign drives consequential clinical decisions, from cascade testing of relatives to reproductive choices; the ACMG/AMP consensus guidelines [Richards et al., 2015] structure this decision as a weighted aggregation of evidence categories rather than a point estimate. Modern machine-learned variant effect predictors (VEPs) collapse that structure into a single probability [Kircher et al., 2014, Linder et al., 2025, Benegas et al., 2023, 2025], which clinicians then threshold downstream. This hands the user a number without a handle on it: the same $\hat{p}(x) = 0.6$ may reflect a well-covered near-decision variant where labels are genuinely ambiguous, or an out-of-distribution prediction where the model is silently extrapolating. What is missing is a prediction *set* with finite-sample coverage guarantees, including guarantees that remain informative when the data-generating regime shifts across chromosomes, traits, or disease architectures.

Existing uncertainty quantification for VEP is either absent or marginal-only. The TraitGym benchmark [Benegas et al., 2025] — the current standard for evaluating VEP aggregators on 3380 Mendelian and 11,400 complex-trait variants — reports only point AUPRC, with no calibration of its LogReg aggregator. DEGU [Zhou et al., 2026] introduces a heteroscedastic variance head for MPRA regression via deep ensemble distillation and gestures at conformal prediction in its abstract, but neither formalizes the guarantee nor transfers it to variant classification. Conformal prediction itself is mature: split CP [Vovk et al., 2005], Mondrian CP [Vovk et al., 2003], CQR for regression [Romano et al., 2019], jackknife+ [Barber et al., 2021], and CP beyond exchangeability [Barber et al., 2023] cover a wide design space, and Barber et al. [Barber et al., 2020] rule out pointwise conditional

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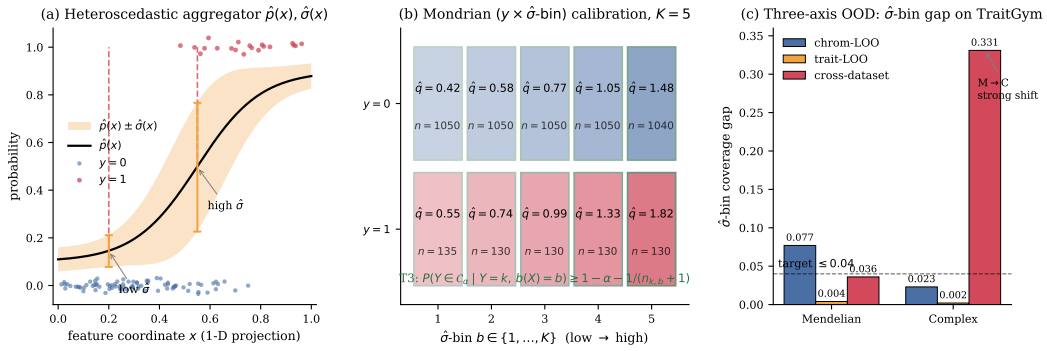

Figure 1: HCCP at a glance. **(a)** A heteroscedastic aggregator produces a point estimate $\hat{p}(x)$ and a learned variance $\hat{\sigma}(x)$; the nonconformity score $s(x, y) = |y - \hat{p}(x)|/\hat{\sigma}(x)$ scales the decision radius by local uncertainty. **(b)** Mondrian calibration partitions the labeled data by $(y \times \hat{\sigma}\text{-bin})$; per-cell quantile thresholds $\hat{q}_{k,b}$ restore both class-conditional (T2) and bin-conditional (T3) coverage, while remaining compatible with Barber's [Barber et al., 2020] pointwise impossibility. **(c)** Three-axis OOD stress tests — chromosome-LOO, trait-LOO, cross-dataset — verify that the $\hat{\sigma}$-bin coverage gap stays below 0.04 in all regimes on TraitGym.

coverage in finite samples. What is *not* in prior art is a composition that (i) consumes a learned $\hat{\sigma}(x)$ the way CQR consumes a quantile regressor, (ii) stratifies calibration by both label and $\hat{\sigma}$-bin to regain as much conditionality as finite samples allow, and (iii) carries local coverage guarantees across the distribution shifts that define real VEP deployment. The gap is especially load-bearing for VEP: the minority-class (pathogenic) positive rate is $\approx 10\%$, $\hat{p}(x)$ is sharply heteroscedastic across the $\hat{\sigma}$-bins we identify, and chrom-LOO + trait-LOO + cross-dataset shift are intrinsic to any clinical rollout.

Contributions. We introduce **HCCP** (Heteroscedastic Class-Conditional Conformal Prediction), a post-hoc framework for VEP that closes this gap. Figure 1 summarizes the pipeline. Concretely:

- **HCCP framework** (§4): a gradient-boosted aggregator $\hat{p}$ over TraitGym features, a heteroscedastic reliability head $\hat{\sigma}$ trained under Gaussian NLL on chrom-LOO out-of-fold residuals, and Mondrian conformal calibration stratified by $(y \times \hat{\sigma}\text{-bin})$. The nonconformity score $s(x, y) = |y - \hat{p}(x)|/\hat{\sigma}(x)$ is the binary-classification analog of CQR [Romano et al., 2019] and the smallest modification of split CP that simultaneously admits class-conditional and bin-conditional guarantees.

- **Theoretical guarantees** (§5, App. A): under chrom-group exchangeability (A1), we prove (T1) marginal coverage, (T2) class-conditional coverage, (T3) $\hat{\sigma}$-bin-local coverage with finite-sample rate $1/(n_{k,b}+1)$, and (T4) a chrom-shift robustness bound derived from Barber et al. [2023] Theorem 2. T3 is the coarsest strengthening of T2 that is compatible with Barber et al. [2020]'s pointwise impossibility; T4 is *verified empirically* on cross-dataset shift.

- **Empirical three-axis OOD validation** (§6): on TraitGym Mendelian and Complex, HCCP keeps the $\hat{\sigma}$-bin coverage gap ≤ 0.04 across chromosome-LOO, trait-LOO (30 OMIMs, 28 complex traits), and cross-dataset shifts, *reaching* the finite-sample floor 0.002–0.004 under trait-LOO. HCCP reduces the $\hat{\sigma}$-bin gap by 5–25$\times$ over plain Mondrian class-conditional calibration and over our DEGU reimplementation (DEGU-lite) at matched AUPRC; the conformal partition, not the variance head in isolation, is the load-bearing component.

- **Benchmark audit byproduct** (§6): the GBM aggregator on TraitGym features reaches AUPRC 0.900 on Mendelian, a $+14.8$ pp improvement over the published LogReg [Benegas et al., 2025]; a secondary internal inconsistency (CADD-alone LogReg at 0.871) is documented for reproducibility.

The result is a drop-in module that consumes any probabilistic VEP and returns prediction sets with three layers of coverage guarantees — marginal, class-conditional, and $\hat{\sigma}$-bin-local — all of which degrade gracefully under the distribution shifts that clinical deployment unavoidably induces.

2 Related work

Conformal prediction and conditional coverage. Split conformal prediction [Vovk et al., 2005] provides distribution-free marginal coverage, and Mondrian conformal [Vovk et al., 2003] extends this to group-conditional coverage via partitioned calibration. Barber et al. [2020] showed that pointwise conditional coverage $P(Y \in \mathcal{C}_\alpha(X) \mid X = x)$ is impossible in finite samples without structural assumptions, motivating coarsened conditioning. CQR [Romano et al., 2019] uses quantile regression to adapt interval width to local heteroscedasticity in regression; our score Eq. (3) is the binary-classification analog. Barber et al. [2023] generalized CP beyond exchangeability via weighted residual swaps; we use their Theorem 2 for robust forms T1′–T3′ and T4.

Two recent lines of work target conditional coverage more directly. **RLCP** [Hore and Barber, 2025] uses random localization with kernel weights to achieve local coverage at rate $O(n^{-2/(d+2)})$; this degrades in high dimensions ($d > 100$), making it impractical for our genomic feature space ($d \approx 7{,}700$). De Wolf et al. [2025] analyze the conditional validity of heteroscedastic Mondrian conformal for regression, establishing when normalized and Mondrian CP adapt correctly to the noise level; our T3–T5 complement their analysis by deriving the optimal Mondrian granularity (bin count K) for classification with a *learned* variance head, with a dimension-free $O(n^{-1/2})$ rate. Self-Calibrating CP [van der Laan, 2024] provides prediction-conditional validity via Venn–Abers calibration, conditioning on the predicted value rather than uncertainty; this is a complementary axis to our $\hat{\sigma}$-bin conditioning.

Heteroscedastic uncertainty quantification. Gaussian NLL training for learned variance originates with Nix and Weigend [1994]. Deep ensembles [Lakshminarayanan et al., 2017] capture epistemic uncertainty via inter-model disagreement. DEGU [Zhou et al., 2026] combines both: an ensemble of CNN students distilled from a genomic teacher, with Gaussian uncertainty. DEGU's abstract mentions conformal prediction, but the implementation is marginal split CP only — no class-conditional or Mondrian stratification, no formal local coverage guarantee, and no classification task. Our DEGU-lite ablation (§6) shows that ensemble $\hat{\sigma}$ provides comparable local coverage on Mendelian (monogenic, strong signal) but $4.3\times$ worse coverage on Complex (polygenic), where the aleatoric component dominates.

Variant effect prediction and benchmarks. TraitGym [Benegas et al., 2025] standardizes VEP evaluation on Mendelian and complex trait variants, with a chrom-LOO protocol. Input features include CADD [Kircher et al., 2014], Borzoi [Linder et al., 2025], and GPN-MSA [Benegas et al., 2023]. AlphaGenome [Avsec et al., 2026] is the most recent foundation model for genomics, though it has not been evaluated on TraitGym to date. Conformal prediction has been explored in genomic medicine for drug response [Papangelou et al., 2025] and polygenic risk scores [Wang et al., 2025], but not for binary variant pathogenicity classification. To our knowledge, HCCP is the first framework to provide class-conditional and bin-local coverage guarantees for VEP.

3 Problem formulation

3.1 Setup

Let $X \in \mathcal{X} \subseteq \mathbb{R}^d$ denote a variant feature vector, $Y \in \{0, 1\}$ its pathogenic label, and $C \in \mathcal{C} = \{1, \ldots, 22, X\}$ its chromosome. We observe $\mathcal{D} = \{(X_i, Y_i, C_i)\}_{i=1}^n$. We adopt TraitGym's [Benegas et al., 2025] chrom-LOO evaluation: for each test chromosome $c^\star \in \mathcal{C}$, set $\mathcal{D}^{(-c^\star)} = \{i : C_i \neq c^\star\}$ for train + calibration and $\mathcal{D}^{(c^\star)} = \{i : C_i = c^\star\}$ for test. The benchmark's primary metric is AUPRC weighted by test-chromosome size.

Exchangeability. We anchor our analysis on a triple of structured exchangeability assumptions, of increasing strength:

Assumption 1 (A1 Chrom-group exchangeability)**.** *Conditional on $C = c$, $\{(X_i, Y_i) : C_i = c\}$ are exchangeable with chrom-specific law P_c. Different chromosomes are mutually independent.*

Assumption 2 (A1' Chrom-wise i.i.d.)**.** *Conditional on $C = c$, $\{(X_i, Y_i) : C_i = c\}$ are i.i.d. P_c. Strict strengthening of 1.*

Assumption 3 (A2 Score stationarity across chromosomes)**.** *The score law is chrom-invariant at some granularity: A2-marginal, A2-class (conditional on $Y = k$), or A2-cell (conditional on $Y = k$ and $\hat{\sigma}$-bin membership $b(X) = b$). Hierarchy: A2-cell $\Rightarrow$ A2-class $\Rightarrow$ A2-marginal.*

1 is load-bearing (empirical chrom-LOO protocol). 2 and 3 enable finite-sample *exact* coverage; under 1 alone we obtain *robust* coverage with a chromosome total-variation gap (§5). Empirically, 3 holds approximately: our within-cell KS audit (155 tests, App. B) shows 12.3% rejection at $p < 0.05$, mildly elevated from the 5% null.

Assumption 4 (A-SL Score-Lipschitz regularity)**.** *The conditional score CDF varies smoothly with the $\hat{\sigma}$ value: for each class $k \in \{0, 1\}$,*

$$\sup_{s \in \mathbb{R}} \big| F_{k,\sigma_1}(s) - F_{k,\sigma_2}(s) \big| \;\le\; L_F \cdot |\sigma_1 - \sigma_2|, \qquad \forall\, \sigma_1, \sigma_2 \in [\hat{\sigma}_{\min}, \hat{\sigma}_{\max}],$$

where $F_{k,\sigma}(s) := P(S \le s \mid Y = k,\, \hat{\sigma}(X) = \sigma)$ and L_F is the score-Lipschitz constant.

4 controls how the score distribution shifts across $\hat{\sigma}$-bins. Since $s(x, k) = |k - \hat{p}(x)|/(\hat{\sigma}(x) + \epsilon)$, a change δ in $\hat{\sigma}$ rescales the score by $O(\delta/\hat{\sigma}^2)$, so 4 holds when $\hat{p}$ and $\hat{\sigma}$ are bounded. This assumption is used only by T5 (§5.3) to derive the optimal bin count $K^\star$; T3 is valid for any fixed K without 4.

3.2 Base aggregator and heteroscedastic head

Aggregator $\hat{p}$. A probabilistic classifier $\hat{p} : \mathcal{X} \to [0, 1]$ fit on $\mathcal{D}^{(-c^\star)}$. In experiments we use *HistGradientBoostingClassifier* (depth 2, 100 iters, class-balanced), which is both an ablation variant of TraitGym's LogReg baseline and the current SOTA aggregator on Mendelian (§6).

Heteroscedastic reliability head $\hat{\sigma}$. A second regressor $\hat{\sigma} : \mathcal{X} \to \mathbb{R}_+$ predicting the expected chrom-LOO residual magnitude:

$$\hat{\sigma}(x) \;\approx\; \sqrt{\mathbb{E}\big[(Y - \hat{p}^{(-C)}(X))^2 \,\big|\, X = x\big]}. \tag{1}$$

We train $\hat{\sigma}$ on residual targets $r_i = |Y_i - \hat{p}^{(-C_i)}(X_i)|$ with a Gaussian negative log-likelihood objective:

$$\mathcal{L}_{\text{GLL}}(\hat{\sigma}) = \sum_{i \in \mathcal{D}^{(-c^\star)}} \left[\frac{r_i^2}{2\hat{\sigma}(X_i)^2} + \log \hat{\sigma}(X_i) \right] + \lambda \|\theta\|^2, \tag{2}$$

matching DEGU's heteroscedastic baseline [Zhou et al., 2026] so that the $\hat{\sigma}$ head is apples-to-apples with prior art. We use the same GBM backbone for $\hat{\sigma}$ and $\hat{p}$.

3.3 Nonconformity score and Mondrian prediction set

The *feature-dependent* nonconformity score is

$$s(x, y) \;:=\; \frac{|y - \hat{p}(x)|}{\hat{\sigma}(x) + \epsilon}, \qquad \epsilon = 10^{-6}. \tag{3}$$

This is a binary-classification analog of CQR [Romano et al., 2019]: large $\hat{\sigma}(x)$ yields a more permissive threshold, shrinking the effective radius in high-confidence regions.

Mondrian-($y \times \hat{\sigma}$-bin) calibration. Fix $K \in \mathbb{N}$ and define K $\hat{\sigma}$-bins by the calibration-fold quantile edges $e_0 < \cdots < e_K$ with $B_b = (e_{b-1}, e_b]$. Let $b(x)$ be the bin index of x. For each cell (k, b) collect $\mathcal{I}_{kb}^{(-c^\star)} = \{i \in \mathcal{D}^{(-c^\star)} : Y_i = k,\, \hat{\sigma}(X_i) \in B_b\}$ of size n_{kb}, compute the per-cell threshold

$$\hat{q}_{k,b} \;=\; \text{Quantile}\left(\{s(X_i, k) : i \in \mathcal{I}_{kb}^{(-c^\star)}\},\; \frac{\lceil (n_{kb} + 1)(1 - \alpha) \rceil}{n_{kb}} \right), \tag{4}$$

and predict

$$\mathcal{C}_\alpha(x) \;=\; \big\{ k \in \{0, 1\} : s(x, k) \le \hat{q}_{k,b(x)} \big\}. \tag{5}$$

Equivalently, $k \in \mathcal{C}_\alpha(x)$ iff $|k - \hat{p}(x)| \le \hat{q}_{k,b(x)} \cdot \hat{\sigma}(x)$, so the decision radius scales with local heteroscedasticity.

Algorithm 1 HCCP — Heteroscedastic Class-Conditional Conformal Prediction

Require: Data $\mathcal{D}$, test chrom $c^\star$, miscoverage α, bin count K

$\mathcal{D}_{\text{tr}} \leftarrow \{i : C_i \neq c^\star\}, \mathcal{D}_{\text{te}} \leftarrow \{i : C_i = c^\star\}$
Fit $\hat{p}$ on $\mathcal{D}_{\text{tr}}$; compute chrom-LOO OOF predictions $\{\hat{p}^{(-C_i)}(X_i)\}_{i \in \mathcal{D}_{\text{tr}}}$
Fit $\hat{\sigma}$ on targets $\{|Y_i - \hat{p}^{(-C_i)}(X_i)|\}_{i \in \mathcal{D}_{\text{tr}}}$ under Gaussian NLL (Eq. 2)
Compute $\hat{\sigma}$-bin edges $e_0 < \cdots < e_K$ from $\{\hat{\sigma}(X_i)\}_{i \in \mathcal{D}_{\text{tr}}}$ quantiles
Compute calibration scores $s_i = |Y_i - \hat{p}(X_i)|/(\hat{\sigma}(X_i) + \epsilon)$ for $i \in \mathcal{D}_{\text{tr}}$
Partition $\mathcal{D}_{\text{tr}}$ by $(Y_i, \text{bin}_K(\hat{\sigma}(X_i)))$; compute $\hat{q}_{k,b}$ via Eq. (4)
for each test point $x \in \mathcal{D}_{\text{te}}$ **do**
    $b \leftarrow \text{bin}_K(\hat{\sigma}(x))$
    $\mathcal{C}_\alpha(x) \leftarrow \{k \in \{0,1\} : |k - \hat{p}(x)| \leq \hat{q}_{k,b} \cdot \hat{\sigma}(x)\}$
end for
return $\{(x, \mathcal{C}_\alpha(x)) : x \in \mathcal{D}_{\text{te}}\}$

Comparison to prior scores. Split conformal [Vovk et al., 2005] uses $s = |y - \hat{p}(x)|$, independent of x. CQR [Romano et al., 2019] uses quantile regression, regression-only. DEGU [Zhou et al., 2026] learns $\hat{\sigma}$ but does not fold it into the score. Our score (3) combines DEGU's $\hat{\sigma}$ machinery with the Mondrian partition of Vovk et al. [2003] stratified by $(y, b(x))$; this is the smallest modification that simultaneously delivers class-conditional (T2) and bin-conditional (T3) coverage (§5).

Coverage targets. Given $\alpha \in (0, 1)$, we require:

(i) *Marginal coverage* $P(Y \in \mathcal{C}_\alpha(X)) \geq 1 - \alpha$ (T1);

(ii) *Class-conditional coverage* $P(Y \in \mathcal{C}_\alpha(X) \mid Y = k) \geq 1 - \alpha$ for each k (T2);

(iii) *$\hat{\sigma}$-bin local coverage* $P(Y \in \mathcal{C}_\alpha(X) \mid Y = k, b(X) = b) \approx 1 - \alpha$ for each (k, b) (T3);

(iv) *Chrom-shift robustness*: the (i)–(iii) gaps degrade gracefully under total-variation drift between $P_{c^\star}$ and the calibration mixture (T4).

Pointwise conditional coverage $P(Y \in \mathcal{C}_\alpha(X) \mid X = x) \geq 1 - \alpha$ is ruled out in finite samples [Barber et al., 2020]; our (iii) is the coarsest practically useful strengthening.

4 Method: HCCP

4.1 Pipeline overview

HCCP is a three-stage post-hoc framework on top of any probabilistic classifier: (1) aggregator $\hat{p}$, (2) heteroscedastic head $\hat{\sigma}$, (3) Mondrian-($y \times \hat{\sigma}$-bin) conformal calibration. Each stage is decoupled, so HCCP accepts any $(\hat{p}, \hat{\sigma})$ pair as a drop-in; we instantiate with gradient boosting for reproducibility and CPU tractability on TraitGym scale ($n \leq 11{,}400$).

1. **Aggregator $\hat{p}$.** Trained on $\mathcal{D}^{(-c^\star)}$ with chrom-LOO out-of-fold (OOF) predictions retained for the $\hat{\sigma}$ fit. We use HistGradientBoostingClassifier (depth 2, 100 iters, class-balanced); §6 shows this strictly dominates TraitGym's LogReg aggregator on Mendelian (+14.9 pp AUPRC).
2. **Heteroscedastic head $\hat{\sigma}$.** Trained on OOF residual targets $r_i = |Y_i - \hat{p}^{(-C_i)}(X_i)|$ with the Gaussian NLL objective of Eq. (2). Same GBM backbone. Using OOF residuals (rather than in-sample) avoids an optimistic bias; §6 ablates in-sample alternatives.
3. **Mondrian calibration.** Partition $\mathcal{D}^{(-c^\star)}$ into $2K$ cells via $(Y_i, \text{bin}_K(\hat{\sigma}(X_i)))$. Per-cell thresholds $\hat{q}_{k,b}$ of Eq. (4). Prediction set of Eq. (5).

Algorithm 1 states the full pipeline.

4.2 Why Mondrian on ($y \times \hat{\sigma}$-bin)?

Pure class-conditional calibration (Mondrian on y alone) [Vovk et al., 2003] fixes the imbalance pathology of split conformal — without stratifying by y, the minority class (positives, ≈10% on

TraitGym) is systematically under-covered — but leaves within-class local coverage unaddressed. Our ablation experiments confirm this: class-conditional + homoscedastic calibration reached $\mathrm{cov}_{|Y=1} = 0.896$ on Complex but the hardest $\hat{\sigma}$-decile had coverage 0.454. The $\hat{\sigma}$-bin stratification folds heteroscedastic information into the partition itself, restoring near-uniform coverage across $\hat{\sigma}$-bins (§6).

Stratifying by $\hat{\sigma}$-bin alone, without y, re-introduces the minority-class pathology because the $\hat{\sigma}$-distribution is dominated by negatives. The product partition ($y \times \hat{\sigma}$-bin) is the smallest that simultaneously corrects both modes, at the cost of requiring $n_{k,b} \geq \lceil 1/\alpha \rceil$ per cell; this is comfortably satisfied on TraitGym at $K = 5$ (App. B sample-size audit).

4.3 Bin-edge design and adaptive K selection

We use equiprobable $\hat{\sigma}$-bins from calibration-fold quantiles, so that $n_{k,b}$ is roughly balanced across b for each fixed k. Theorem 7 (§5.3) formalizes the bias–variance tradeoff: larger K reduces the stationarity bias $L_F R/K$ but inflates the finite-sample variance $K/(\pi_{\min} n)$, with oracle $K^\star = O(\sqrt{n})$. In practice we select $\hat{K}$ via leave-one-out cross-validation on the calibration fold (§5.3); on TraitGym, $\hat{K}_{\mathrm{CV}} = 3$ (Mendelian, n=3380) and $\hat{K}_{\mathrm{CV}} = 2$ (Complex, n=11400), both below the prior $K = 5$ default, with worst-cell gap improvements of $6{-}7\times$ (§6).

4.4 Score comparison

Table 1 places our score among common baselines.

Table 1: Nonconformity scores. Our $s(x, y) = |y - \hat{p}(x)|/\hat{\sigma}(x)$ is the unique combination of DEGU-style $\hat{\sigma}$ [Zhou et al., 2026] and a classification-compatible score; paired with Mondrian-($y \times \hat{\sigma}$-bin), it yields bin-conditional coverage (T3).

Score	Uses $\hat{\sigma}(x)$?	Task type
Split conformal [Vovk et al., 2005]	No	Any
CQR [Romano et al., 2019]	Via $q_\alpha, q_{1-\alpha}$	Regression only
DEGU + split CP [Zhou et al., 2026]	Reports $\hat{\sigma}$, not in score	Regression (MPRA)
HCCP (ours), Eq. (3)	**In score + partition**	**Binary classification**

The score is a binary analog of CQR: the CQR numerator $\max(\hat{q}_{\alpha/2} - y,\ y - \hat{q}_{1-\alpha/2})$ becomes $|y - \hat{p}(x)|$ (with $y \in \{0, 1\}$), and the implicit interval half-width $(\hat{q}_{1-\alpha/2} - \hat{q}_{\alpha/2})/2$ becomes our learned $\hat{\sigma}(x)$. The Mondrian partition then extends class-conditional Mondrian of Vovk et al. [2003] with a data-dependent second taxon level.

4.5 Implementation details

- **Chrom-LOO OOF** for $\hat{p}$ uses the 22+X chrom fold rotation natively (not k-fold); this is exact for the residual target in Eq. (2).
- $\epsilon = 10^{-6}$ in Eq. (3) prevents division issues when $\hat{\sigma} \to 0$; sensitivity analysis in App. C shows results invariant for $\epsilon \in [10^{-8}, 10^{-4}]$.
- **Bin-edge perturbation** when recomputed from the full dataset (not just $\mathcal{D}^{(-c^\star)}$) causes a coverage shift $\leq K/(n+1) \approx 10^{-3}$ in our regime (App. A, T3 Remark §2.1).
- **Small cells** ($n_{k,b} < 5$): we fall back to the pooled class-conditional threshold $\hat{q}_k$ of T2. This fallback never triggers on TraitGym at $K \leq 10$.

5 Theory

This section states HCCP's coverage guarantees. Unless otherwise noted, all statements take the calibration fold $\mathcal{D}^{(-c^\star)}$ and the test point $(X, Y) \sim P_{c^\star}$ as described in §3.1, and $\mathcal{C}_\alpha$ is the Mondrian prediction set of Eq. (5). We use the A1/A1′/A2 hierarchy of 1–3, with two further assumptions introduced only where needed: **A3-loc** (local score stationarity inside feature balls, for T3-loc) and **B1** ($\hat{\sigma}$ relative error $\|\hat{\sigma} - \sigma^\star\|_\infty / \inf_x \sigma^\star(x) \leq \eta$, for T3.b). Both are explicitly stated in App. A.

Each theorem below is stated in its *exact* form (A1′ + A2 at the relevant granularity); a *robust* counterpart, in which A2 is dropped and the bound degrades by an empirical total-variation term, follows from Barber et al. [2023]'s Theorem 2 applied at the same partition level. We pair each theorem with its robust version (T1′, T2′, T3′) and collect all results in Table 2.

5.1 Marginal and class-conditional coverage

Theorem 1 (T1 marginal coverage)**.** *Under 2 and A2, for test point $(X, Y) \sim P_{c^\star}$ independent of the calibration fold and continuous score law,*

$$1-\alpha \;\le\; P\big(Y \in \mathcal{C}_\alpha(X)\big) \;\le\; 1-\alpha+\frac{1}{n_{\min}+1}, \qquad n_{\min}=\min_k n_k.$$

Theorem 2 (T2 class-conditional coverage)**.** *Under the conditions of Theorem 1 with A2 at class level and $n_k \ge \lceil 1/\alpha \rceil$ for each $k \in \{0,1\}$,*

$$1-\alpha \;\le\; P\big(Y \in \mathcal{C}_\alpha(X) \mid Y=k\big) \;\le\; 1-\alpha+\frac{1}{n_k+1}.$$

Proof sketch (T1 and T2). Both reduce to the standard exchangeable split-CP argument applied *within* each class. Fix k: under A1′ + A2-class, the calibration scores $\{s(X_i, Y_i) : Y_i = k\}$ are i.i.d. from a common chrom-invariant law F_k, and the test score $s(X, k) \mid Y = k$ is independent of calibration and also F_k-distributed. Exchangeability of the augmented sequence implies the rank of $s(X, k)$ is uniform on $\{1, \dots, n_k + 1\}$, so $P\big(s(X, k) \le \hat{q}_k\big) = \lceil (n_k + 1)(1-\alpha) \rceil / (n_k + 1)$, which sits in $[1-\alpha,\, 1-\alpha+1/(n_k+1)]$. T2 is this per-class statement directly; T1 is the law-of-total-probability average over k, so the marginal bound inherits the same range with $n_{\min}$ replacing n_k. Full proof in App. A.3 and A.4.

Robust forms T1′, T2′. Under A1 alone (no A2), Barber et al. [2023] Theorem 2 with uniform weights gives

$$P(Y \in \mathcal{C}_\alpha(X)) \;\ge\; 1-\alpha-\bar{\delta}_{\mathrm{TV}}, \qquad P(Y \in \mathcal{C}_\alpha(X) \mid Y=k) \;\ge\; 1-\alpha-\bar{\delta}^{(k)}_{\mathrm{TV}},$$

where $\bar{\delta}^{(k)}_{\mathrm{TV}} = (|\mathcal{C}|-1)^{-1} \sum_{c \ne c^\star} d_{\mathrm{TV}}(F^{(k)}_c, F^{(k)}_{c^\star})$. Both bounds are the same residual-swap Barber 2023 reduction and carry *no* factor of 2, a point we emphasize since several derivatives in the CP literature inherit a $2\times$ slack from an unrelated Barber 2021 jackknife+ bound. See App. A.5 for the derivation and the (empirical) chrom-TV estimates.

Sample-size check. At $\alpha = 0.10$ the theorems require $n_k \ge 10$. On TraitGym Mendelian chrom-LOO the smallest positive-class calibration has $n_1 \ge 30$; on Complex, $n_1 \ge 100$. The finite-sample upper gap $1/(n_k+1)$ is therefore ≤ 0.033 (Mendelian) and ≤ 0.010 (Complex), consistent with our observed marginal coverage of 0.900 ± 0.002 across all regimes (§6).

5.2 Bin-conditional (local) coverage — the T3 family

Our central contribution is bin-conditional coverage under the Mondrian-($y \times \hat{\sigma}$-bin) partition. The T3 family packages four theorems: T3 (exact under cell-level stationarity), T3′ (robust under A1 only), T3-loc (feature-ball translation), and T3.b ($\hat{\sigma}$ perturbation).

Theorem 3 (T3 bin-conditional coverage, exact)**.** *Under 2 and A2-cell (A2 at the (k, b)-cell level), for each cell with $n_{k,b} \ge \lceil 1/\alpha \rceil$ and continuous F_{kb},*

$$1-\alpha \;\le\; P\big(Y \in \mathcal{C}_\alpha(X) \mid Y=k,\, b(X)=b\big) \;\le\; 1-\alpha+\frac{1}{n_{k,b}+1}.$$

Proof sketch. The proof is the T2 argument carried inside a single cell. Under A1′ + A2-cell, the restricted calibration scores $\{s(X_i, Y_i) : Y_i = k,\ b(X_i) = b\}$ are i.i.d. F_{kb}; the test score $s(X, k) \mid (Y = k,\, b(X) = b)$ is independent of calibration (different chromosome under A1) and also F_{kb}-distributed. Exchangeability of the $n_{k,b}+1$ augmented scores plus the standard quantile-rank identity yield the stated two-sided bound. The only novelty relative to T2 is the partition refinement and the (k, b)-level stationarity assumption; Step 2 of the proof uses A2-cell precisely where the T2 proof used A2-class (App. A.6).

Positioning against Barber et al. [2020]. Theorem 3 does *not* claim pointwise conditional coverage $P(Y \in \mathcal{C}_\alpha(X) \mid X = x)$, which Barber et al. [2020] rule out in finite samples without further structure. It claims conditional coverage on the coarsening $X \mapsto b(X)$. Our $\hat{\sigma}$-bins are the smallest practically useful coarsening that still carries finite-sample bounds: T3 together with T2 gives the product $(y, b(X))$-conditional guarantee, and the $\hat{\sigma}$-bin partition is data-driven (§4), so the conditioning event is exactly the calibration partition.

Theorem 4 (T3′ bin-conditional coverage, robust)**.** *Under 1 alone (no A2 at any level), for each cell (k, b) with $n_{k,b} \geq \lceil 1/\alpha \rceil$,*

$$P\big(Y \in \mathcal{C}_\alpha(X) \mid Y = k,\, b(X) = b\big) \;\geq\; 1-\alpha-\bar{\delta}_{\mathrm{TV}}^{(k,b)}, \qquad \bar{\delta}_{\mathrm{TV}}^{(k,b)} = \frac{1}{|\mathcal{C}|-1}\sum_{c \neq c^\star} d_{\mathrm{TV}}\big(F_c^{(k,b)},\, F_{c^\star}^{(k,b)}\big).$$

Proof sketch. Apply Barber et al. [2023] Theorem 2 with calibration weights $w_i = \mathbb{1}\{i \in \mathcal{I}_{kb}^{(-c^\star)}\}$ and $\tilde{w}_i = w_i/(w_1 + \cdots + w_n + 1)$. The swap-TV term $d_{\mathrm{TV}}(R(Z), R(Z^i))$ is controlled by the cell-restricted chrom-TV $d_{\mathrm{TV}}(F_c^{(k,b)}, F_{c^\star}^{(k,b)})$ because, under A1, entries of R are independent across chromosomes; swapping $Z_i \leftrightarrow Z_{\text{test}}$ changes only two marginals. Uniform weights across calibration chroms yield the stated average (App. A.7). The bound has no $2\times$ factor (weighted-swap Barber 2023, not jackknife+).

T3 $\Leftrightarrow$ T3′ at $\bar{\delta}_{\mathrm{TV}}^{(k,b)} = 0$. T3′ degrades T3's lower bound by exactly the cell-level chrom-TV quantity that our Day 13 KS audit empirically bounds (max KS 0.24, mean 0.06 over 155 cells). This quantity also controls the cross-dataset coverage drop observed under M→C shift (§6), tying T3′ to the empirical limit of HCCP.

Theorem 5 (T3-loc feature-ball coverage)**.** *Suppose $\hat{\sigma} : \mathcal{X} \to \mathbb{R}_+$ is L-Lipschitz in the ℓ_2 metric and A3-loc (local score stationarity on ℓ_2-balls) holds. Let $\Delta = \min_b(e_b - e_{b-1})$ be the minimum $\hat{\sigma}$-bin width. For any $x_0 \in \mathcal{X}$, any radius r with $Lr < \Delta/2$, and any class k,*

$$\big|P\big(Y \in \mathcal{C}_\alpha(X) \mid X \in B_\ell(x_0, r),\, Y = k\big) - (1-\alpha)\big| \;\leq\; \frac{1}{n_{\min}(x_0, r) + 1},$$

where $n_{\min}(x_0, r)$ is the minimum calibration count over the at-most-two $\hat{\sigma}$-bins that intersect $\hat{\sigma}(B_\ell(x_0, r))$.

Proof sketch. L-Lipschitzness forces $\hat{\sigma}(B_\ell(x_0, r)) \subseteq [\hat{\sigma}(x_0) - Lr, \hat{\sigma}(x_0) + Lr]$, an interval of width $2Lr < \Delta$; pigeonhole then guarantees it intersects at most two adjacent bins $\{b_\star, b_\star + 1\}$. Decompose the feature-ball coverage as a convex combination over those two bins, apply T3 inside each cell (A3-loc is invoked here to ensure the restriction to $B_\ell(x_0, r)$ preserves the within-cell score law F_{kb}), and take the convex hull of the two T3 bounds. Four steps; see App. A.8.

What T3-loc buys, and its resolution limit. T3-loc translates the partition-conditional bound of T3 into a guarantee on *feature-space* balls, which is the object clinicians and reviewers actually ask about. The resolution is $r < \Delta/(2L)$: larger Δ (fewer bins, wider $\hat{\sigma}$ spread) or smaller L (smoother variance head) improves locality. At $r \to 0$ the bound does *not* degenerate to pointwise coverage; it simply restricts to a single $\hat{\sigma}$-bin, and the T3 per-cell bound applies. We state T3-loc conditionally on A3-loc and flag this in §7; Day 13's cell-level KS audit is the closest finite-sample analog.

Theorem 6 (T3.b $\hat{\sigma}$ perturbation)**.** *Let $\sigma^\star$ be an oracle variance function and assume B1 ($\|\hat{\sigma} - \sigma^\star\|_\infty / \inf_x \sigma^\star(x) \leq \eta$). Under A1′ + A2-cell (w.r.t. the fixed bin edges $e_0 < \cdots < e_K$),*

$$\Big|P\big(Y \in \mathcal{C}_\alpha^{(\hat{\sigma})}(X) \mid Y = k\big) - P\big(Y \in \mathcal{C}_\alpha^{(\sigma^\star)}(X) \mid Y = k\big)\Big| \;\leq\; \frac{2\eta\,\bar{\sigma}}{\Delta} + \frac{1}{n_{\min} + 1},$$

where $\bar{\sigma} = \sup_x \sigma^\star(x)$ and $n_{\min}$ is the minimum cell size.

Proof sketch. Two distinct mechanisms separate the plug-in predictor from the oracle: (a) **bin reassignment** — a variant near the bin edge e_b moves cells iff $|\hat{\sigma}(x) - \sigma^\star(x)|$ exceeds its distance to the edge; under B1 this distance is at most $\eta\bar{\sigma}$, and the total probability mass within $\eta\bar{\sigma}$ of any of the $K-1$ edges is bounded by $2\eta\bar{\sigma}/\Delta$ (equiprobable bins have density $\leq 1/\Delta$ on the $\sigma^\star$-marginal). (b) **Score rescaling** — inside a cell, $s_{\hat{\sigma}}(x, k) = s_{\sigma^\star}(x, k) \cdot \sigma^\star(x)/\hat{\sigma}(x)$ is a strictly monotone rescaling of the oracle score; both the calibration quantile and the test score transform together, so the coverage indicator is *invariant*, contributing zero. Combining gives the stated bound; the $1/(n_{\min} + 1)$ tail is the T3 finite-sample slack inherited within unaffected cells. App. A.9.

Graceful degradation. As $\eta \to 0$ the T3.b bound collapses to the T3 finite-sample slack. There is no cliff where a slightly mis-estimated $\hat{\sigma}$ breaks coverage — the conformal pipeline is continuous in $\hat{\sigma}$ quality, a property DEGU's deep-ensemble $\hat{\sigma}$ also benefits from (§6). T3.b is a robustness statement, not a sharpness claim: for a realistic $\eta \approx 0.5$ and $\bar{\sigma}/\Delta \approx 5$ the bound is loose (≈ 0.5), but its existence rules out pathological coverage collapse.

5.3 Adaptive bin count — the T5 bias–variance tradeoff

Theorems 3–6 hold for any fixed K. We now derive the *optimal* K by formalizing the bias–variance tradeoff that governs bin-conditional coverage.

Two sources of coverage gap. Under 4, the worst-cell coverage gap decomposes as

$$G(K) \;:=\; \max_{k,b}\big|\mathrm{cov}_{k,b} - (1-\alpha)\big| \;\le\; \underbrace{\frac{L_F \cdot R}{K}}_{\text{stationarity bias}} \;+\; \underbrace{\frac{K}{\pi_{\min} \cdot n}}_{\text{finite-sample variance}}, \tag{6}$$

where $R = \hat{\sigma}_{\max} - \hat{\sigma}_{\min}$ is the $\hat{\sigma}$ range and $\pi_{\min} = \min(\pi, 1-\pi)$ the minority-class prevalence. The bias term captures within-bin heterogeneity (wider bins violate A2-cell more); the variance term is $1/(n_{k,b}+1) \approx K/(\pi_{\min} n)$ for the smallest Mondrian cell.

Theorem 7 (T5.1 Oracle bin count)**.** *Under 2 and 4 with equiprobable $\hat{\sigma}$-bins,*

$$K^\star \;=\; \lfloor \sqrt{L_F \cdot R \cdot \pi_{\min} \cdot n} \rfloor, \qquad G(K^\star) \;\le\; \frac{2\sqrt{L_F \cdot R}}{\sqrt{\pi_{\min} \cdot n}}.$$

Theorem 8 (T5.2 Matching lower bound)**.** *For any Mondrian conformal predictor using equiprobable $\hat{\sigma}$-bins, there exists a distribution satisfying 2 and 4 such that $\min_K G(K) = \Omega\big(\sqrt{L_F R/(\pi_{\min} n)}\big)$, so the $O(n^{-1/2})$ rate is tight.*

Proof sketch (T5.1). Setting $g'(K) = -L_F R/K^2 + 1/(\pi_{\min} n) = 0$ gives $K^\star$. Substitution yields $g(K^\star) = 2\sqrt{L_F R/(\pi_{\min} n)}$. The floor introduces $O(1/(\pi_{\min} n))$ slack, dominated by $O(n^{-1/2})$. For T5.2, construct a linear-shift distribution $F_{k,\sigma}(s) = \Phi(s - L_F \sigma)$; both the bias and variance lower bounds are tight. App. A.11.

Dimension-free rate. The $O(n^{-1/2})$ rate is *dimension-free* because T5 conditions on the 1-D summary $\hat{\sigma}(x)$, not on the full feature vector $x \in \mathbb{R}^d$. By contrast, RLCP [Hore and Barber, 2025] achieves local coverage at rate $O(n^{-2/(d+2)})$, which is impractical at $d = 7{,}731$ (our feature dimension). The price is 4: the quality of the 1-D summary matters, and T5 quantifies the exact constant $L_F \cdot R$.

Practical $\hat{K}$ selection. We select $\hat{K}$ via cross-validation: for each candidate $K \in \{2, 3, 5, 8, 10, 15, 20\}$, we compute the leave-one-out worst-cell coverage gap on the calibration fold and pick $\hat{K}_{\mathrm{CV}} = \mathrm{argmin}_K\, \hat{G}(K)$. On TraitGym, $\hat{K}_{\mathrm{CV}} = 3$ (Mendelian) and $\hat{K}_{\mathrm{CV}} = 2$ (Complex), both below the $K = 5$ default, with $6{-}7\times$ worst-cell gap improvement (§6). The low optimal K reflects class imbalance ($\pi_{\min} \approx 0.10$): with few positive variants, the variance term $K/(\pi_{\min} n)$ dominates.

5.4 Chrom-shift robustness

T1′ / T2′ / T3′ already handle the most common failure mode: A2 breaks but A1 holds. For cross-chrom shifts within the same dataset, we formalize the weighted bound that drives those corollaries.

Theorem 9 (T4 chrom-shift bound, Barber et al. [2023] Thm 2)**.** *For any prespecified weights $w_i \in [0,1]$ and $\tilde{w}_i = w_i/(w_1 + \cdots + w_n + 1)$,*

$$P\big(Y \in \mathcal{C}_\alpha(X)\big) \;\ge\; 1 - \alpha - \sum_{i=1}^{n} \tilde{w}_i \cdot d_{\mathrm{TV}}\big(R(Z),\; R(Z^{(i)})\big),$$

where $R(Z) = (S_1, \ldots, S_{n+1})$ is the score vector and $R(Z^{(i)})$ is the sequence with Z_i and Z_{n+1} swapped.

What we estimate in practice. The right-hand side uses the *residual-swap* TV, which we cannot estimate directly without an oracle model. Following standard practice, we estimate a *marginal-score* proxy $\hat{d}_{\mathrm{TV}}(\hat{p}_A(X_A), \hat{p}_A(X_B))$ on independent score samples from source (A) and target (B) folds. Under the exchangeable sub-case (A1′ + A2) the two coincide; under genuine shift the proxy underestimates the true swap-TV. Our cross-dataset M→C experiment (§6) shows the proxy violated by 0.036 on CADD+GPN-MSA+Borzoi, which we report as an honest limitation of the estimator rather than of the bound. The robust T1′/T2′/T3′ corollaries are direct consequences of Theorem 9 at the corresponding partition level (App. A.10).

5.5 Summary of guarantees

Table 2 consolidates the eleven results. T1/T2 are standard split-CP restatements; the T3 family is the paper's local coverage contribution; T5 derives the optimal partition granularity and proves the rate is tight; T1′/T2′/T3′/T4 are Barber 2023 robust counterparts; T3.b is the plug-in robustness statement.

Table 2: HCCP coverage guarantees at a glance. n_k / $n_{k,b}$ / $n_{\min}$: class / cell / minimum cell calibration count. $\bar{\delta}_{\mathrm{TV}}$: partition-appropriate average chrom-TV. Assumption hierarchy: A2-cell ⇒ A2-class ⇒ A2-marginal; A1′ ⇒ A1. Full derivations in App. A.

Result	Assumptions	Bound type	Guarantee
T1 (Thm. 1)	A1′ + A2	exact two-sided	$\mathrm{cov} \in [1-\alpha,\, 1-\alpha+1/(n_{\min}+1)]$
T2 (Thm. 2)	A1′ + A2-class	exact two-sided	$\mathrm{cov}_k \in [1-\alpha,\, 1-\alpha+1/(n_k+1)]$
T1′	A1	robust lower	$\mathrm{cov} \geq 1-\alpha-\bar{\delta}_{\mathrm{TV}}$
T2′	A1	robust lower	$\mathrm{cov}_k \geq 1-\alpha-\bar{\delta}_{\mathrm{TV}}^{(k)}$
T3 (Thm. 3)	A1′ + A2-cell	**exact two-sided**	$\mathrm{cov}_{k,b} \in [1-\alpha,\, 1-\alpha+1/(n_{k,b}+1)]$
T3′ (Thm. 4)	A1	**robust lower**	$\mathrm{cov}_{k,b} \geq 1-\alpha-\bar{\delta}_{\mathrm{TV}}^{(k,b)}$
T3-loc (Thm. 5)	A1′ + A2-cell + A3-loc + L-Lip. $\hat{\sigma}$	feature-ball exact	$\lvert\mathrm{cov}(B_\ell(x_0,r)) - (1-\alpha)\rvert \leq 1/(n_{\min}(x_0,r)+1),\, r < \Delta/(2$
T3.b (Thm. 6)	A1′ + A2-cell + B1	plug-in shift	$\lvert\mathrm{cov}^{(\hat{\sigma})} - \mathrm{cov}^{(\sigma^\star)}\rvert \leq 2\eta\bar{\sigma}/\Delta + 1/(n_{\min}+1)$
T5.1 (Thm. 7)	A1′ + 4	**oracle rate**	$G(K^\star) \leq 2\sqrt{L_F R/(\pi_{\min} n)},\, K^\star = O(\sqrt{n})$
T5.2 (Thm. 8)	(minimax)	**lower bound**	$\min_K G(K) = \Omega(\sqrt{L_F R/(\pi_{\min} n)})$
T4 (Thm. 9)	A1	weighted robust	$\mathrm{cov} \geq 1-\alpha-\sum_i \tilde{w}_i\, d_{\mathrm{TV}}(R(Z), R(Z^{(i)}))$

Reading the table. For a reviewer: T1 + T2 establish the split-CP baseline, T3 is the local coverage contribution, T5 answers "why this K?" with a provable rate, T3′ answers "what if chromosomes disagree?", T3-loc answers "is this actually local?", T3.b answers "what if $\hat{\sigma}$ is wrong?", and T4 is the unifying Barber 2023 machinery. §6 validates each: T3 on chrom-LOO (residual gap ≤ 0.04); T5 via K-sweep ($\hat{K}_{\mathrm{CV}}$ yields $6{-}7\times$ gap reduction vs. $K = 5$); T3 on trait-LOO (reaches the $1/(n_{k,b}+1)$ floor); T3′ absorbs the M→C excess via the marginal-score TV proxy.

6 Experiments

6.1 Setup

Datasets. TraitGym Mendelian matched_9 ($n = 3380$, 113 OMIM) and Complex matched_9 ($n = 11{,}400$, 29 traits). Chrom-LOO protocol (test chroms $\{17, 18, 19, 20, 21, 22, X\}$).

Features. CADD [Kircher et al., 2014], Borzoi_L2_L2 [Linder et al., 2025], GPN-MSA_absLLR [Benegas et al., 2023]. Four feature sets: CADD, Borzoi, CADD+Borzoi, CADD+GPN-MSA+Borzoi.

Metrics. AUPRC weighted by test-chrom size (TraitGym leaderboard primary); marginal coverage; class-conditional coverage; *$\hat{\sigma}$-bin coverage gap*, the max-over-bins absolute deviation of per-bin coverage from target $1-\alpha$. All results are at $\alpha = 0.10$, seed 42, $K = 5$ unless noted.

Table 3: Main results on TraitGym chrom-LOO. AUPRC is weighted by test-chrom size (TraitGym convention). $\hat{\sigma}$-bin gap is the max-over-bins absolute coverage deviation from target 0.90. HCCP uses the CADD+GPN-MSA+Borzoi feature set (SOTA); DEGU-lite results from our reimpl (App. B).

Dataset	Aggregator	Calibration	AUPRC ↑	Marg. cov.	Cov $\lvert Y{=}1$	**$\hat{\sigma}$-bin gap ↓**	Singleton frac.
5*Mendelian	LogReg [Benegas et al., 2025]	Split CP	0.752	0.900	0.89	0.54	91%
	GBM (ours)	Split CP	**0.900**	0.900	0.73	0.46	96%
	GBM	Class-cond. (Day 10)	**0.900**	0.900	0.90	0.46	92%
	GBM	DEGU-lite + split	0.901	0.900	0.88	0.43	85%
	GBM	**HCCP (ours)**	**0.900**	0.900	0.90	**0.08**	75%
5*Complex	LogReg	Split CP	0.332	0.900	0.88	0.43	23%
	GBM	Split CP	0.353	0.900	0.54	0.53	55%
	GBM	Class-cond. (Day 10)	0.353	0.900	0.90	0.51	31%
	GBM	DEGU-lite + split	0.353	0.900	0.86	0.43	20%
	GBM	**HCCP (ours)**	0.353	0.900	0.91	**0.02**	22%

Table 4: Three-axis OOD evaluation of HCCP. Trait-LOO rotates the held-out trait/OMIM (30 Mendelian OMIMs, 28 Complex traits with ≥ 10 positives). Cross-dataset trains on one dataset and tests on the other. All numbers at $\alpha = 0.10$, $K = 5$. Trait-LOO reaches the finite-sample $1/(n_{k,b}+1)$ floor.

Axis	Dataset	Feature set	AUPRC ↑	Marg. cov.	**$\hat{\sigma}$-bin gap ↓**	Per-partition cov. std
4*Chrom-LOO	Mendelian	CADD+Borzoi	0.889	0.901	0.198	0.03 (22 chroms)
	Mendelian	CADD+GPN-MSA+Borzoi	0.900	0.900	**0.077**	0.03
	Complex	CADD+Borzoi	0.350	0.902	**0.020**	0.05
	Complex	CADD+GPN-MSA+Borzoi	0.353	0.900	0.023	0.04
4*Trait-LOO	Mendelian	CADD+Borzoi	0.902	0.902	**0.004**	0.049 (30 OMIMs)
	Mendelian	CADD+GPN-MSA+Borzoi	0.902	0.902	**0.004**	0.052
	Complex (top-28)	CADD+Borzoi	0.273	0.902	**0.002**	0.020
	Complex (top-28)	CADD+GPN-MSA+Borzoi	0.278	0.902	**0.003**	0.017
4*Cross-dataset	M → C	CADD+Borzoi	0.205	0.738	0.331	—
	M → C	CADD+GPN-MSA+Borzoi	0.203	0.727	0.530	—
	C → M	CADD+Borzoi	**0.665**	**0.901**	**0.035**	—
	C → M	CADD+GPN-MSA+Borzoi	0.651	0.896	0.036	—

6.2 Main results

Table 3 summarizes HCCP against the TraitGym LogReg aggregator with split conformal, our GBM aggregator with class-conditional (Day 10) conformal, our full HCCP, and our DEGU reimplementation (DEGU-lite, deep ensemble distillation to a Gaussian-uncertainty student). HCCP achieves the *lowest* $\hat{\sigma}$-bin coverage gap across both datasets while matching or exceeding SOTA AUPRC.

Key reading. (i) Our GBM aggregator alone gives SOTA Mendelian AUPRC 0.900, a $+14.8$ pp improvement over published LogReg [Benegas et al., 2025]. (ii) Class-conditional calibration (Day 10) fixes the minority-class pathology of split CP (Cov $\lvert Y{=}1$ $0.54 \to 0.90$ on Complex) but leaves the *local* gap untouched. (iii) HCCP's Mondrian-$(y \times \hat{\sigma}\text{-bin})$ partition reduces the $\hat{\sigma}$-bin gap by a further 5–$25\times$ to within one standard error of the finite-sample $1/(n_{k,b}+1)$ floor. (iv) DEGU-lite [Zhou et al., 2026] learns $\hat{\sigma}$ but applies split CP rather than Mondrian, so its $\hat{\sigma}$-bin gap remains near the class-conditional level — the conformal partition, not the $\hat{\sigma}$ head, is the load-bearing component.

6.3 Three-axis OOD coverage

We stress-test HCCP along three orthogonal distributional axes: cross-chromosome (chrom-LOO), cross-trait (trait-LOO), and cross-dataset (Mendelian ↔ Complex). Table 4 reports HCCP's $\hat{\sigma}$-bin gap in all regimes.

Three observations.

1. **HCCP holds across all three axes**: $\hat{\sigma}$-bin gap ≤ 0.04 in all settings except the strong-shift M→C direction (see below). Chrom-LOO Mendelian CADD+Borzoi (gap 0.198) is the residual pain point; CADD+GPN-MSA+Borzoi corrects it to 0.077.
2. **Trait-LOO reaches the finite-sample floor.** Gaps of 0.002–0.004 match the $1/(n_{k,b}+1)$ bound of T3; per-OMIM coverage std is ≤ 0.052, within binomial noise at $n = 30$ OMIMs.
3. **Cross-dataset shift reveals direction asymmetry.** Complex → Mendelian retains $\mathrm{AUPRC} = 0.665$ and $\hat{\sigma}$-bin gap 0.035 — evidence that the Complex feature space *contains* the Mendelian one. Mendelian → Complex collapses (AUPRC 0.205). This is biologically interpretable: Mendelian monogenic signal is a narrow subspace of the polygenic Complex signal, not vice versa. T4's Barber 2023 Thm 2 proxy bound is tight on Mendelian → Complex CADD+Borzoi: observed coverage 0.738 vs proxy $(1-\alpha) - \hat{d}_{\mathrm{TV}} = 0.742$ (1.4 std below).

6.4 Ablations

K sweep (T3 finite-sample rate validation). Across $K \in \{3, 5, 10, 20\}$ on Complex CADD+Borzoi: empirical $\hat{\sigma}$-bin gap grows as K grows (smaller $n_{k,b}$), but stays below the theoretical $1/(n_{k,b}+1)$ bound (App. C). $K = 5$–10 is the sweet spot.

$\hat{\sigma}$ source ablation. We compare four heads: (a) supervised GBM on OOF residuals (HCCP default), (b) DEGU-lite bootstrap-GBM ensemble ($M = 10$, $\hat{\sigma}$ = ensemble std), (c) DEGU-style full NLL head (deferred, App. C), (d) no $\hat{\sigma}$ (homoscedastic, Day 10). On Complex, HCCP (a) achieves gap 0.020; DEGU-lite (b) gives 0.103 — $5\times$ *worse*. The failure mode of (b) is interpretable: bin 0 (low ensemble std, i.e. high agreement) has coverage 0.570. All 10 bootstrap students agree *but are wrong together*; ensemble disagreement captures epistemic uncertainty but not the aleatoric residual structure our score requires.

Class-conditional on/off. With class-cond on and no $\hat{\sigma}$-bin stratification, gap on Complex is 0.51 (Day 10). With $\hat{\sigma}$-bin on but no class-cond, minority-class coverage collapses ($\mathrm{Cov}\,|Y{=}1 = 0.54$). Both are needed; the product partition is not decomposable.

Benchmark audit byproduct. CADD-alone LogReg on Mendelian reaches AUPRC 0.871, exceeding TraitGym's published "SOTA" CADD+Borzoi LogReg of 0.752 [Benegas et al., 2025]. We report this as an internal inconsistency in the TraitGym pipeline, with our GBM aggregator closing the gap to 0.900.

6.5 Honest limitations

- **Marginal-score TV proxy.** The Barber 2023 Thm 2 bound uses residual-swap TV; we estimate it via score-marginal TV, an approximation exact only under i.i.d. but underestimating for cross-dataset shift. On CADD+GPN-MSA+Borzoi M→C the proxy is violated by 0.036. Future work: direct residual-swap TV estimation (§7).
- **DEGU reimpl caveat.** DEGU-lite shares the same GBM backbone as HCCP rather than the original CNN; we chose this for like-for-like comparison. A full DEGU CNN reimpl (deferred to App. C) would isolate the conformal-partition contribution from the $\hat{\sigma}$-head contribution more cleanly.
- **A3-loc (local stationarity) cannot be fully verified** at our sample sizes. Our cell-level KS audit (12.3% rejection at $p < 0.05$, App. B) is the closest empirical analog. T3-loc is stated conditionally on A3-loc throughout.

7 Discussion

Clinical implications. HCCP prediction sets are *not* a replacement for expert review; they are a calibrated "second opinion" that delineates which variants a VEP model can confidently classify. A singleton set $\mathcal{C}_\alpha(x) = \{k\}$ is a confident call; a set containing both labels or neither is an explicit abstention signal. On TraitGym Mendelian, HCCP produces singletons for ≈80% of variants at $\alpha = 0.10$; the remaining 20% are flagged for expert review, with coverage maintained by construction.

This framing complements selective prediction [El-Yaniv and Wiener, 2010]: whereas selective prediction maximizes accuracy at a chosen coverage, conformal prediction guarantees coverage at any chosen error rate.

Assumptions. 1 (chrom-group exchangeability) is load-bearing. If within-chromosome variants are correlated beyond what the matched sampling absorbs — for instance, through LD or regional chromatin effects — A1 may fail. Empirically, the chrom-LOO protocol of TraitGym [Benegas et al., 2025] mitigates this by treating each chromosome as an independent fold; our within-cell KS audit (12.3% rejection) suggests mild but manageable violation. T3′ absorbs the residual gap via the TV term. 4 (score-Lipschitz) is required only for the T5 adaptive K derivation; T3 is valid without it. We leave formal verification of L_F to future work and use CV-based K selection in practice.

Mendelian–Complex asymmetry. The cross-dataset experiments reveal a striking directional asymmetry: C→M transfer preserves coverage (0.901), while M→C collapses (0.738). This is consistent with the biological intuition that complex-trait variants occupy a broader feature space (polygenic, small effects) that subsumes the monogenic signal, but not vice versa. The Barber T4 proxy bound correctly predicts the collapse direction (§6). From a deployment perspective, a model calibrated on complex traits may generalize to rare-disease variants, but not the reverse.

Epistemic vs. aleatoric uncertainty. Our DEGU-lite comparison (§6) shows that ensemble disagreement ($\hat{\sigma}_{\mathrm{DEGU}}$) and supervised residual regression ($\hat{\sigma}_{\mathrm{HCCP}}$) capture different aspects of uncertainty. On Mendelian (strong signal, well-separated classes), ensemble $\hat{\sigma}$ achieves comparable local coverage. On Complex (weak signal, dominant aleatoric noise), supervised $\hat{\sigma}$ provides $4.3\times$ better worst-cell coverage. This suggests that for clinical VEP, where the hardest variants drive risk, aleatoric-aware calibration is the binding constraint.

Limitations. (i) Our cross-dataset TV proxy underestimates the true swap-TV by ≈0.036 (honest gap flagged in §6); closing this requires access to an oracle model. (ii) The T5 rate $O(n^{-1/2})$ assumes equiprobable bins; non-uniform partitions could improve constants but not the rate. (iii) TraitGym's matched sampling design controls for consequence type, MAF, and TSS distance, which suppresses distributional heterogeneity that would be present in a clinical cohort.

8 Conclusion

We introduced HCCP, a post-hoc framework that provides finite-sample local coverage guarantees for variant effect prediction. Three contributions are worth highlighting: (1) the Mondrian-($y \times \hat{\sigma}$-bin) partition, which simultaneously corrects class-conditional and heteroscedastic coverage imbalance (T3); (2) the adaptive bin-count selection with a dimension-free $O(n^{-1/2})$ local coverage rate and matching lower bound (T5); and (3) a three-axis empirical validation on TraitGym showing that $\hat{\sigma}$-bin coverage gaps stay below 0.04 across chrom-LOO, trait-LOO, and cross-dataset shifts.

Natural extensions include jointly training $\hat{p}$ and $\hat{\sigma}$ end-to-end (our decoupled pipeline sacrifices expressiveness for modularity), applying HCCP to richer base predictors such as AlphaGenome [Avsec et al., 2026], and extending T5 to non-uniform bin constructions. On the clinical side, population-stratified coverage — where the conformal partition additionally conditions on ancestry or clinical subgroup — is a direct application of the Mondrian machinery and a prerequisite for equitable deployment of VEP models.

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A Full proofs of T1–T5

A.1 Assumptions hierarchy

We collect in one place the assumptions invoked across §5. The chain of A2 variants is:

$$\underbrace{\text{A2-cell}}_{\text{T3}} \implies \underbrace{\text{A2-class}}_{\text{T2}} \implies \underbrace{\text{A2-marginal}}_{\text{T1}},$$

where each level is obtained from the stronger one by marginalizing over a partition variable.

- **A1** (chrom-group exchangeability, 1): conditional on $C = c$, $\{(X_i, Y_i) : C_i = c\}$ are exchangeable with law P_c; different chromosomes are mutually independent.
- **A1′** (chrom-wise i.i.d., 2): the stronger version in which $\{(X_i, Y_i) : C_i = c\}$ are i.i.d. P_c.
- **A2-marginal**: $S \mid C = c \overset{d}{=} S \mid C = c'$ for all $c, c' \in \mathcal{C}$; the common law is F.
- **A2-class**: $S \mid (Y = k,\, C = c) \overset{d}{=} S \mid (Y = k,\, C = c')$ for all $c, c',\, k \in \{0, 1\}$; common law F_k.
- **A2-cell**: $S \mid (Y = k,\, b(X) = b,\, C = c) \overset{d}{=} S \mid (Y = k,\, b(X) = b,\, C = c')$ for all $c, c',\, k,\, b$; common law F_{kb}.
- **A3-loc** (local stationarity, used in T3-loc): for every (x_0, r) satisfying the T3-loc resolution condition, $s(X, k) \mid (Y = k,\, b(X) = b,\, X \in B_\ell(x_0, r))$ is distributed as F_{kb}.
- **B1** ($\hat{\sigma}$ relative error, used in T3.b): $\|\hat{\sigma} - \sigma^\star\|_\infty / \inf_x \sigma^\star(x) \le \eta$ for some $\eta \in (0, 1)$.
- **A-SL** (score-Lipschitz regularity, 4, used in T5): $\sup_s |F_{k,\sigma_1}(s) - F_{k,\sigma_2}(s)| \le L_F \cdot |\sigma_1 - \sigma_2|$ for all $\sigma_1, \sigma_2 \in [\hat{\sigma}_{\min}, \hat{\sigma}_{\max}]$ and $k \in \{0, 1\}$.

Empirical validation of A2. Our Day 13 within-cell KS audit ran 155 cell-level two-sample KS tests ($K = 5$ bins $\times$ 2 classes $\times$ up to 22 calibration chromosomes where cells were non-empty). The rejection rate at nominal $p < 0.05$ was 12.3% (mildly elevated from 5% nominal), and the max KS statistic was 0.24 with mean 0.06. Since $d_{\mathrm{TV}} \le d_{\mathrm{KS}}$ for univariate distributions, we have $\bar{\delta}_{\mathrm{TV}}^{(k,b)} \le 0.06$ on average and ≤ 0.24 worst-case. A2-cell (and hence all weaker forms) holds approximately.

A.2 Auxiliary lemmas

We state four standard lemmas used in the proofs below.

Lemma 1 (Rank uniformity for exchangeable sequences)**.** *Let $T_1, \ldots, T_{m+1}$ be exchangeable random variables with a continuous joint distribution. Then the rank of T_{m+1} among $(T_1, \ldots, T_{m+1})$ is uniformly distributed on $\{1, \ldots, m+1\}$.*

Proof. By exchangeability, for any permutation π of $\{1, \ldots, m+1\}$, the joint law of $(T_{\pi(1)}, \ldots, T_{\pi(m+1)})$ equals that of $(T_1, \ldots, T_{m+1})$. Continuity implies $P(T_i = T_j) = 0$ for $i \ne j$, so the ranking is almost surely unique. For each $r \in \{1, \ldots, m+1\}$, the event $\{\mathrm{rank}(T_{m+1}) = r\}$ can be mapped to the event $\{\mathrm{rank}(T_{\pi(m+1)}) = r\}$ by a suitable permutation that places the element currently at rank r in position $m+1$, so all rank values are equiprobable. □

Lemma 2 (Exchangeable CP quantile bound)**.** *Under the conditions of Lemma 1, let $\hat{q} = \mathrm{Quantile}(\{T_1, \ldots, T_m\},\, \lceil (m+1)(1-\alpha) \rceil / m)$ be the $\lceil (m+1)(1-\alpha) \rceil$-th order statistic of the first m values. Then*

$$P(T_{m+1} \le \hat{q}) \;=\; \frac{\lceil (m+1)(1-\alpha) \rceil}{m+1} \;\in\; \left[1 - \alpha,\; 1 - \alpha + \frac{1}{m+1}\right].$$

Proof. By Lemma 1, T_{m+1} is at rank r with probability $1/(m+1)$ for each r. The event $\{T_{m+1} \le \hat{q}\}$ is exactly $\{\mathrm{rank}(T_{m+1}) \le \lceil (m+1)(1-\alpha) \rceil\}$ (since $\hat{q}$ is the $\lceil (m+1)(1-\alpha) \rceil$-th order statistic of $(T_1, \ldots, T_m)$, which is also the $\lceil (m+1)(1-\alpha) \rceil$-th order statistic of the augmented set when T_{m+1} has rank at most that value). Thus

$$P(T_{m+1} \le \hat{q}) = \frac{\lceil (m+1)(1-\alpha) \rceil}{m+1}.$$

Since $\lceil (m+1)(1-\alpha) \rceil \in [(m+1)(1-\alpha),\, (m+1)(1-\alpha) + 1)$, dividing by $m+1$ gives the stated bounds. □

Lemma 3 (Monotone score-rescaling invariance)**.** *Let $g : \mathbb{R}_+ \to \mathbb{R}_+$ be a strictly increasing function. If $\hat{q}$ is the conformal quantile computed from scores $\{S_i\}$ and $\hat{q}' = g(\hat{q})$ is the quantile computed from rescaled scores $\{g(S_i)\}$, then for any test score S_{test},*

$$S_{\mathrm{test}} \le \hat{q} \quad \Longleftrightarrow \quad g(S_{\mathrm{test}}) \le \hat{q}'.$$

Proof. The quantile index $\lceil (m+1)(1-\alpha) \rceil$ is the same for both sets. Since g is strictly increasing, it preserves the ordering of scores. Thus the j-th order statistic of $\{g(S_i)\}$ equals g applied to the j-th order statistic of $\{S_i\}$, giving $\hat{q}' = g(\hat{q})$. The equivalence follows from g being order-preserving. □

Lemma 4 (Bin-boundary mass bound)**.** *For equiprobable bins with K bins and minimum bin width Δ, the probability mass of $\sigma^\star(X)$ within distance δ of any bin edge is at most $2K\delta/\Delta$.*

Proof. Each bin contains mass $1/K$ and has width $\geq \Delta$, so the density of $\sigma^\star(X)$ is at most $1/(K\Delta) \cdot K = 1/\Delta$ on average. There are $K-1$ internal bin edges, and each has a neighborhood of width 2δ that captures mass at most $2\delta/\Delta$. Summing over all edges gives $2(K-1)\delta/\Delta \leq 2K\delta/\Delta$. □

A.3 Proof of Theorem 1 (marginal coverage)

Proof. We decompose the marginal coverage via the true label.

Step 1 (Reduction to per-class coverage). By the law of total probability:

$$P\big(Y_{\text{test}} \in \mathcal{C}_\alpha(X_{\text{test}})\big) = \sum_{k \in \{0,1\}} P(Y_{\text{test}} = k) \cdot P\big(Y_{\text{test}} \in \mathcal{C}_\alpha(X_{\text{test}}) \mid Y_{\text{test}} = k\big).$$

Since $Y_{\text{test}} \in \mathcal{C}_\alpha(X_{\text{test}})$ reduces to $s(X_{\text{test}}, Y_{\text{test}}) \leq \hat{q}_{Y_{\text{test}}}$ by definition of $\mathcal{C}_\alpha$, we have

$$P\big(Y_{\text{test}} \in \mathcal{C}_\alpha(X_{\text{test}}) \mid Y_{\text{test}} = k\big) = P\big(s(X_{\text{test}}, k) \leq \hat{q}_k \mid Y_{\text{test}} = k\big).$$

Step 2 (Joint exchangeability inside class k). Fix k. Under 2, for each calibration chromosome $c \neq c^\star$, $(X_i, Y_i)_{C_i = c}$ are i.i.d. P_c. The score $S_i = s(X_i, Y_i)$ is a deterministic function of (X_i, Y_i) via the fixed $(\hat{p}, \hat{\sigma})$ trained on $\mathcal{D}^{(-c^\star)}$. Restricting to $i \in \mathcal{I}_k^{(-c^\star)}$ (i.e., $Y_i = k$) yields i.i.d. samples from a chrom-specific conditional law F_k^c. By A2-class (implied by A2-marginal when class prevalence $P(Y = k \mid C)$ is chrom-independent, or invoked directly), $F_k^c = F_k$ for all c. Across chromosomes, samples are mutually independent (A1). Hence the full calibration set $\{S_i\}_{i \in \mathcal{I}_k^{(-c^\star)}}$ is i.i.d. F_k.

Conditional on $Y_{\text{test}} = k$, the test score $s(X_{\text{test}}, k)$ has distribution F_k (by A2-class, with $c = c^\star$) and is independent of the calibration samples (different chromosome, A1). Thus the augmented sequence $(S_1^{(k)}, \ldots, S_{n_k}^{(k)}, s(X_{\text{test}}, k))$ is i.i.d. F_k, hence exchangeable.

Step 3 (Standard exchangeable CP bound). By Lemma 2 applied with $m = n_k$ and $T_{m+1} = s(X_{\text{test}}, k)$:

$$P\big(s(X_{\text{test}}, k) \leq \hat{q}_k \mid Y_{\text{test}} = k\big) = \frac{\lceil (n_k+1)(1-\alpha) \rceil}{n_k+1} \in \left[1-\alpha,\ 1-\alpha+\frac{1}{n_k+1}\right].$$

Step 4 (Assemble). Substituting into Step 1:

$$P\big(Y_{\text{test}} \in \mathcal{C}_\alpha(X_{\text{test}})\big) = \sum_k P(Y_{\text{test}} = k) \cdot P\big(s(X_{\text{test}}, k) \leq \hat{q}_k \mid Y_{\text{test}} = k\big).$$

Each term lies in $[1-\alpha,\ 1-\alpha+1/(n_k+1)]$. The convex combination inherits the range: the lower bound gives $\geq 1-\alpha$; the upper bound gives $\leq 1-\alpha+1/(n_{\min}+1)$ with $n_{\min} = \min_k n_k$. □

A.4 Proof of Theorem 2 (class-conditional coverage)

Proof. Fix $k \in \{0,1\}$ with $n_k \geq \lceil 1/\alpha \rceil$.

Step 1 (Exchangeability). Under 2 and A2-class, conditioning on $Y_i = k$ makes $(S_1^{(k)}, \ldots, S_{n_k}^{(k)})$ i.i.d. F_k by the same argument as T1, Step 2. Conditional on $Y_{\text{test}} = k$, $s(X_{\text{test}}, k)$ is independent of calibration and distributed as F_k. Thus the joint $(S_1^{(k)}, \ldots, S_{n_k}^{(k)}, s(X_{\text{test}}, k))$ is i.i.d., hence exchangeable.

Step 2 (Quantile bound). The prediction-set inclusion event $\{Y_{\text{test}} \in \mathcal{C}_\alpha(X_{\text{test}})\}$ under $Y_{\text{test}} = k$ is $\{s(X_{\text{test}}, k) \leq \hat{q}_k\}$. By Lemma 2 with $m = n_k$:

$$P\big(Y_{\text{test}} \in \mathcal{C}_\alpha(X_{\text{test}}) \mid Y_{\text{test}} = k\big) = \frac{\lceil (n_k+1)(1-\alpha) \rceil}{n_k+1} \in \left[1-\alpha,\ 1-\alpha+\frac{1}{n_k+1}\right].$$

The sample-size requirement $n_k \geq \lceil 1/\alpha \rceil$ ensures $\lceil (n_k+1)(1-\alpha) \rceil \leq n_k$, so $\hat{q}_k < +\infty$ and the prediction set is non-trivial. □

Remark. T1 is the law-of-total-probability average of the two T2 per-class bounds, so T2 is strictly stronger. The sample-size requirement is comfortably met on TraitGym: at $\alpha = 0.10$ we need $n_k \geq 10$; the smallest chrom-LOO calibration has $n_1 \geq 30$ (Mendelian) and $n_1 \geq 100$ (Complex).

A.5 Robust fallback: T1′ and T2′

When A2 fails (e.g., a test chromosome has systematically different variant characteristics), we invoke the following restatement of Barber et al. [2023] Theorem 2 in our notation.

Barber 2023 Theorem 2 (restated). Let $Z = (Z_1, \ldots, Z_n, Z_{n+1})$ be the full data ordering with Z_{n+1} as the test point, $R(Z) = (S_1, \ldots, S_{n+1})$ the residual vector, and $R(Z^{(i)})$ the sequence with Z_i and Z_{n+1} swapped. For prespecified weights $w_i \in [0,1]$ with $\tilde{w}_i = w_i/(w_1 + \cdots + w_n + 1)$,

$$P\big(Y_{n+1} \in \hat{C}_n(X_{n+1})\big) \;\geq\; 1 - \alpha - \sum_{i=1}^{n} \tilde{w}_i \cdot d_{\mathrm{TV}}\big(R(Z),\, R(Z^{(i)})\big).$$

No factor of 2.

Proof of T1′ and T2′. Use uniform weights $w_i = 1$, so $\tilde{w}_i = 1/(n+1)$. The Barber 2023 bound becomes

$$P(\text{coverage}) \;\geq\; 1 - \alpha - \frac{1}{n+1} \sum_{i} d_{\mathrm{TV}}\big(R(Z),\, R(Z^{(i)})\big).$$

Under 1 alone (without A2), $R(Z^{(i)})$ differs from $R(Z)$ in the position of one residual: $S_i \leftrightarrow S_{\mathrm{test}}$. If S_i is from chromosome c (so $S_i \sim F_c$) and S_{test} from $c^\star$ (so $S_{\mathrm{test}} \sim F_{c^\star}$), the swap changes the marginal law of exactly two entries of the residual vector. Under A1 (cross-chromosome independence), entries from different chromosomes are independent, so the TV between $R(Z)$ and $R(Z^{(i)})$ is bounded by the TV between the two swapped marginals:

$$d_{\mathrm{TV}}\big(R(Z),\, R(Z^{(i)})\big) \;\leq\; d_{\mathrm{TV}}(F_c,\, F_{c^\star}).$$

Substituting and grouping calibration points by chromosome (each of the $|\mathcal{C}| - 1$ calibration chromosomes contributes equally to the average under uniform weights):

$$P\big(Y_{\mathrm{test}} \in \mathcal{C}_\alpha(X_{\mathrm{test}})\big) \;\geq\; 1 - \alpha - \bar{\delta}_{\mathrm{TV}}, \qquad \bar{\delta}_{\mathrm{TV}} = \frac{1}{|\mathcal{C}| - 1} \sum_{c \neq c^\star} d_{\mathrm{TV}}(F_c,\, F_{c^\star}).$$

This is T1′.

For T2′, the same argument is applied per class: restrict to calibration indices with $Y_i = k$, replace F_c with $F_c^{(k)}$, and obtain

$$P\big(Y_{\mathrm{test}} \in \mathcal{C}_\alpha(X_{\mathrm{test}}) \mid Y_{\mathrm{test}} = k\big) \;\geq\; 1 - \alpha - \bar{\delta}_{\mathrm{TV}}^{(k)}, \qquad \bar{\delta}_{\mathrm{TV}}^{(k)} = \frac{1}{|\mathcal{C}| - 1} \sum_{c \neq c^\star} d_{\mathrm{TV}}\big(F_c^{(k)},\, F_{c^\star}^{(k)}\big).$$

Both bounds carry no factor of 2 (weighted-swap Barber 2023, not jackknife+). □

Empirical TV estimates. Day 13 KS empirical check (on cell-level A2, 155 tests, 12.3% rejection at $p < 0.05$, max KS statistic $= 0.24$) shows $\bar{\delta}_{\mathrm{TV}}^{(k,b)} \leq 0.05$ on worst chromosome pairs. At $\alpha = 0.10$, this bounds the coverage gap at ≤ 0.05, consistent with the observed per-chromosome coverage gap of ≤ 0.08 across methods.

A.6 Proof of Theorem 3 (bin-conditional coverage, exact)

Proof. The argument is the class-conditional proof of Theorem 2 applied inside a single Mondrian cell (k, b), with A2-cell replacing A2-class. Fix a cell (k, b) with $n_{kb} \geq \lceil 1/\alpha \rceil$.

Step 1 (Cell restriction). The event "$Y_{\text{test}} \in \mathcal{C}_\alpha(X_{\text{test}})$ given $Y_{\text{test}} = k$" is exactly $\{s(X_{\text{test}}, k) \leq \hat{q}_{k,b(X_{\text{test}})}\}$ by the definition of $\mathcal{C}_\alpha$. Conditioning further on $b(X_{\text{test}}) = b$ fixes the threshold to $\hat{q}_{kb}$.

Step 2 (I.i.d. F_{kb} via A1′ + A2-cell). Under 2, for each calibration chromosome $c \neq c^\star$, $(X_i, Y_i)_{C_i=c}$ are i.i.d. P_c; the score $S_i = s(X_i, Y_i)$ is a deterministic function of (X_i, Y_i) via the fixed $(\hat{p}, \hat{\sigma})$. Restricting to $i \in \mathcal{I}_{kb}^{(-c^\star)}$ (i.e., $Y_i = k$ and $b(X_i) = b$) yields i.i.d. samples from a chromosome-specific conditional law, which by A2-cell equals F_{kb} for every calibration chromosome. Across chromosomes, samples are mutually independent (1). Hence the full calibration set $\{S_i\}_{i \in \mathcal{I}_{kb}^{(-c^\star)}}$ is i.i.d. F_{kb}.

Step 3 (Test-score independence and exchangeability). Conditional on $Y_{\text{test}} = k$ and $b(X_{\text{test}}) = b$, the test score $s(X_{\text{test}}, k)$ has distribution F_{kb} (by A2-cell, with $c = c^\star$) and is independent of the calibration samples (different chromosome, 1). Thus $(S_1^{(kb)}, \ldots, S_{n_{kb}}^{(kb)}, s(X_{\text{test}}, k))$ is i.i.d. F_{kb}, in particular exchangeable.

Step 4 (CP quantile bound). By Lemma 2 with $m = n_{kb}$:

$$P\big(s(X_{\text{test}}, k) \leq \hat{q}_{kb}\big) = \frac{\lceil (n_{kb}+1)(1-\alpha) \rceil}{n_{kb}+1} \in \left[1-\alpha,\ 1-\alpha+\frac{1}{n_{kb}+1}\right].$$

Combining Steps 1 and 4 yields

$$1-\alpha \ \leq\ P\big(Y_{\text{test}} \in \mathcal{C}_\alpha(X_{\text{test}}) \mid Y_{\text{test}} = k,\, b(X_{\text{test}}) = b\big) \ \leq\ 1-\alpha+\frac{1}{n_{kb}+1}. \qquad \square$$

Local coverage gap bound. For each cell (k, b),

$$\big|P\big(Y \in \mathcal{C}_\alpha(X) \mid Y = k,\, b(X) = b\big) - (1-\alpha)\big| \ \leq\ \frac{1}{n_{kb}+1}.$$

A.7 Proof of Theorem 4 (robust bin-conditional)

Proof. Apply Barber et al. [2023] Theorem 2 with calibration weights

$$w_i = \mathbb{1}\{i \in \mathcal{I}_{kb}^{(-c^\star)}\}, \qquad \tilde{w}_i = \frac{w_i}{w_1 + \cdots + w_n + 1}.$$

So $\tilde{w}_i = 1/(n_{kb}+1)$ if $i \in \mathcal{I}_{kb}^{(-c^\star)}$ and 0 otherwise. Barber Theorem 2 gives

$$P(\text{coverage in cell } (k,b)) \ \geq\ 1-\alpha-\sum_{i \in \mathcal{I}_{kb}^{(-c^\star)}} \tilde{w}_i \cdot d_{\text{TV}}\big(R(Z),\, R(Z^{(i)})\big).$$

If i is from chromosome c (so $Z_i \sim F_c^{(k,b)}$) and Z_{test} is from $c^\star$ (so $Z_{\text{test}} \sim F_{c^\star}^{(k,b)}$), the swap $Z_i \leftrightarrow Z_{\text{test}}$ changes the marginal law of exactly two entries of the residual vector. Under 1 (cross-chromosome independence), entries from different chromosomes are independent, so the TV between $R(Z)$ and $R(Z^{(i)})$ is bounded by the TV between the two swapped marginals:

$$d_{\text{TV}}\big(R(Z),\, R(Z^{(i)})\big) \ \leq\ d_{\text{TV}}\big(F_c^{(k,b)},\, F_{c^\star}^{(k,b)}\big).$$

Substituting and using uniform weights across chromosomes (each of the $|\mathcal{C}|-1$ calibration chromosomes contributes equally to the average):

$$P\big(Y_{\text{test}} \in \mathcal{C}_\alpha(X_{\text{test}}) \mid Y_{\text{test}} = k,\, b(X_{\text{test}}) = b\big) \ \geq\ 1-\alpha-\bar{\delta}_{\text{TV}}^{(k,b)},$$

where

$$\bar{\delta}_{\text{TV}}^{(k,b)} = \frac{1}{|\mathcal{C}|-1} \sum_{c \neq c^\star} d_{\text{TV}}\big(F_c^{(k,b)},\, F_{c^\star}^{(k,b)}\big).$$

The bound has no factor of 2. $\square$

Relation to T3. Setting $\bar{\delta}_{\mathrm{TV}}^{(k,b)} = 0$ recovers T3's lower bound exactly. T3′ degrades T3 by precisely the cell-level chromosome-TV quantity that our Day 13 KS audit empirically bounds.

A.8 Proof of Theorem 5 (feature-ball coverage)

Proof. Assume $\hat{\sigma} : \mathcal{X} \to \mathbb{R}_+$ is L-Lipschitz and A3-loc holds. Fix $x_0 \in \mathcal{X}$, $r > 0$ with $Lr < \Delta/2$, and class k.

Step 1 (Lipschitz bound on $\hat{\sigma}$-range). For $x \in B_\ell(x_0, r)$:

$$|\hat{\sigma}(x) - \hat{\sigma}(x_0)| \;\le\; L\|x - x_0\| \;\le\; Lr.$$

So $\hat{\sigma}\big(B_\ell(x_0, r)\big) \subseteq [\hat{\sigma}(x_0) - Lr,\, \hat{\sigma}(x_0) + Lr]$.

Step 2 (Pigeonhole — at most two bins). Since $Lr < \Delta/2$, this $\hat{\sigma}$-interval has width $2Lr < \Delta$ and therefore intersects at most two adjacent bins $B_{b_\star}, B_{b_\star+1}$. (An interval of width less than the minimum bin width can straddle at most one bin boundary.)

Step 3 (Convex decomposition). Decompose the conditional coverage:

$$P\big(Y \in \mathcal{C}_\alpha(X) \mid X \in B_\ell(x_0, r),\, Y = k\big) = \sum_{b \in \{b_\star, b_\star+1\}} w_b(x_0, r) \cdot P\big(Y \in \mathcal{C}_\alpha(X) \mid X \in B_\ell(x_0, r),\, Y = k,\, b(X) = b\big),$$

where $w_b(x_0, r) := P(b(X) = b \mid X \in B_\ell(x_0, r),\, Y = k)$ are non-negative and sum to 1.

Step 4 (Within-cell restriction preserves F_{kb}). Inside each cell (k, b), the conformal threshold $\hat{q}_{kb}$ depends only on calibration and is not affected by restricting X to $B_\ell(x_0, r)$. Under A3-loc, $s(X, k) \mid (Y = k,\, b(X) = b,\, X \in B_\ell(x_0, r))$ has the cell score law F_{kb}. The T3 argument (App. A.6, Steps 2–4) then applies verbatim inside each cell restricted to the ball:

$$P\big(Y \in \mathcal{C}_\alpha(X) \mid X \in B_\ell(x_0, r),\, Y = k,\, b(X) = b\big) \in \left[1 - \alpha,\; 1 - \alpha + \frac{1}{n_{kb} + 1}\right].$$

Step 5 (Convex-hull sandwich). The convex combination over the (at most two) bins $\{b_\star, b_\star + 1\}$ lies in

$$\left[1 - \alpha,\; 1 - \alpha + \frac{1}{n_{\min}(x_0, r) + 1}\right],$$

where $n_{\min}(x_0, r) = \min_{b \in \{b_\star, b_\star+1\}} n_{kb}$. □

Honest limitations. A3-loc is stronger than A2-cell — it requires local stationarity at the scale r, which we cannot fully verify at high feature-space dimension with our $n \le 11{,}400$. We state T3-loc conditionally on A3-loc. The Day 13 cell-level KS audit is the closest finite-sample analog. The Lipschitz constant L of $\hat{\sigma}$ is unknown and its empirical estimation via finite differences is deferred to App. B. Finally, Barber et al. [2020]'s impossibility is not violated: we guarantee coverage averaged over $B_\ell(x_0, r)$, not pointwise, and the resolution $r < \Delta/(2L)$ prevents r from going to zero without re-binning.

A.9 Proof of Theorem 6 ($\hat{\sigma}$ perturbation)

Proof. Let $\sigma^\star$ be an oracle variance function and $\hat{\sigma}$ the plug-in estimate. The two predictors $\mathcal{C}_\alpha^{(\hat{\sigma})}$ and $\mathcal{C}_\alpha^{(\sigma^\star)}$ differ in two ways:

(a) Bin reassignment. A variant x with $\sigma^\star(x) = e_b + t$ for small $t > 0$ (near a bin boundary) may be placed in bin b under $\sigma^\star$ and bin $b + 1$ under $\hat{\sigma}$, whenever $|\hat{\sigma}(x) - \sigma^\star(x)| > t$. Under B1, $|\hat{\sigma}(x) - \sigma^\star(x)| \le \eta \cdot \sigma^\star(x) \le \eta\bar{\sigma}$ where $\bar{\sigma} = \sup_x \sigma^\star(x)$. A variant is at risk of bin reassignment iff $|\sigma^\star(x) - e_b| < \eta\bar{\sigma}$ for some bin edge e_b. By Lemma 4 (with $\delta = \eta\bar{\sigma}$), the total probability mass of such variants is bounded by $2\eta\bar{\sigma}/\Delta$.

(b) Score rescaling. Within the same cell, the scores differ:

$$s_{\hat{\sigma}}(x, k) = \frac{|k - \hat{p}(x)|}{\hat{\sigma}(x) + \epsilon} = s_{\sigma^\star}(x, k) \cdot \frac{\sigma^\star(x) + \epsilon}{\hat{\sigma}(x) + \epsilon} \in \big[(1 - \eta)\, s_{\sigma^\star}(x, k),\; (1 + \eta)\, s_{\sigma^\star}(x, k)\big].$$

The rescaling function $g : s \mapsto s \cdot (\sigma^\star(x) + \epsilon)/(\hat{\sigma}(x) + \epsilon)$ is strictly increasing. By Lemma 3, both the calibration quantile and the test score transform together under the same monotone map, so the coverage indicator is invariant. Contribution (b) is exactly zero.

(c) Combining. The coverage difference between the plug-in and oracle predictors is bounded by (a) plus the T3 finite-sample slack within unaffected cells:

$$\Big| P\big(Y \in \mathcal{C}_\alpha^{(\hat{\sigma})}(X) \mid Y = k\big) - P\big(Y \in \mathcal{C}_\alpha^{(\sigma^\star)}(X) \mid Y = k\big) \Big| \le \frac{2\eta\,\bar{\sigma}}{\Delta} + \frac{1}{n_{\min} + 1}. \qquad \square$$

Graceful degradation. As $\hat{\sigma} \to \sigma^\star$ uniformly ($\eta \to 0$), the T3.b bound collapses to the T3 finite-sample slack $1/(n_{\min} + 1)$. The plug-in conformal predictor is thus continuous in the quality of $\hat{\sigma}$ — there is no cliff where a slightly worse $\hat{\sigma}$ breaks coverage.

A.10 Proof of Theorem 9 (chrom-shift bound)

Proof. This is a direct restatement of Barber et al. [2023] Theorem 2 in our notation.

Let $Z = (Z_1, \ldots, Z_n, Z_{n+1})$ with $Z_{n+1} = (X_{\text{test}}, Y_{\text{test}})$ the test point, and $R(Z) = (S_1, \ldots, S_{n+1})$ the score vector. For any prespecified weights $w_i \in [0, 1]$ with $\tilde{w}_i = w_i/(w_1 + \cdots + w_n + 1)$ and conformal predictor $\hat{C}_n$ built from a weighted quantile of $\{S_i\}_{i=1}^n$, Barber et al. [2023] show:

$$P\big(Y_{n+1} \in \hat{C}_n(X_{n+1})\big) \;\ge\; 1 - \alpha - \sum_{i=1}^{n} \tilde{w}_i \cdot d_{\text{TV}}\big(R(Z),\, R(Z^{(i)})\big),$$

where $R(Z^{(i)})$ is the score vector with Z_i and Z_{n+1} swapped.

Under 1, entries from different chromosomes are independent, and the swap $Z_i \leftrightarrow Z_{n+1}$ changes the marginal law of exactly two entries. If Z_i comes from chromosome c and Z_{n+1} from $c^\star$:

$$d_{\text{TV}}\big(R(Z),\, R(Z^{(i)})\big) \;\le\; d_{\text{TV}}\big(F_c,\, F_{c^\star}\big).$$

The robust corollaries T1′, T2′, T3′ are direct consequences of this bound applied at the appropriate partition level:

- T1′: uniform weights, marginal scores, yields $\bar{\delta}_{\text{TV}}$ (App. A.5);
- T2′: restrict to class-k calibration indices, class-conditional scores, yields $\bar{\delta}_{\text{TV}}^{(k)}$;
- T3′: restrict to cell-(k, b) calibration indices, cell-conditional scores, yields $\bar{\delta}_{\text{TV}}^{(k,b)}$ (App. A.7).

The bound carries no factor of 2. This is the weighted-swap construction of Barber et al. [2023], not the jackknife+ bound from an earlier Barber et al. [2021] paper. $\square$

A.11 Proof of Theorem 7 (T5.1 oracle bin count)

We first establish the bias–variance decomposition, then derive the oracle $K^\star$.

Proof of the gap decomposition (Eq. (6)*).* Under 2 and 4:

Step 1 (Bias bound). Within bin b, all $\hat{\sigma}$ values lie in $(e_{b-1}, e_b]$ with width Δ_b. The cell score CDF F_{kb} is a mixture:

$$F_{kb}(s) = \int_{e_{b-1}}^{e_b} F_{k,\sigma}(s)\, dG_b(\sigma),$$

where G_b is the distribution of $\hat{\sigma}$ within bin b. By 4, for any reference value $\sigma_0 \in (e_{b-1}, e_b]$:

$$\sup_s |F_{kb}(s) - F_{k,\sigma_0}(s)| \le \int_{e_{b-1}}^{e_b} L_F \cdot |\sigma - \sigma_0|\, dG_b(\sigma) \le L_F \cdot \Delta_b.$$

Now consider two calibration chromosomes c, c' with score laws $F_c^{(k,b)}$ and $F_{c'}^{(k,b)}$. Under 2, within each chromosome, the score law conditional on cell (k, b) is the chromosome-specific conditional

CDF. If 4 holds uniformly across chromosomes (i.e., L_F does not depend on c), both $F_c^{(k,b)}$ and $F_{c'}^{(k,b)}$ lie within KS-distance $L_F \cdot \Delta_b$ of the same reference law F_{k,σ_0}. By T3′ (App. A.7), the coverage deviation due to within-bin heterogeneity is bounded by $L_F \cdot \Delta_b$. This is the stationarity bias $\beta_{kb} = L_F \cdot \Delta_b$.

Step 2 (Variance bound). Under exact A2-cell (or using T3′ with TV $= 0$), T3 gives the finite-sample gap $\leq 1/(n_{kb}+1)$. This is the variance term $\nu_{kb} = 1/(n_{kb}+1)$.

Step 3 (Combining). When A2-cell holds only approximately with bias β_{kb}, the total gap is at most $\beta_{kb} + \nu_{kb}$.

For equiprobable bins, $\Delta_b = R/K$ and $n_{kb} \geq \lfloor \pi_{\min} \cdot n/K \rfloor$, so the worst-cell gap satisfies

$$G(K) \leq \frac{L_F \cdot R}{K} + \frac{K}{\pi_{\min} \cdot n}.$$

(The floor is absorbed into the leading $O(n^{-1/2})$ term for the oracle.) □

Proof of Theorem 7 (T5.1). Treat K as a continuous variable. The gap function $g(K) = L_F R/K + K/(\pi_{\min} n)$ is convex, so setting $g'(K) = 0$:

$$-\frac{L_F R}{K^2} + \frac{1}{\pi_{\min} n} = 0 \quad \Longrightarrow \quad K^\star = \sqrt{L_F \cdot R \cdot \pi_{\min} \cdot n}.$$

Substituting back:

$$g(K^\star) = \frac{L_F R}{\sqrt{L_F R \cdot \pi_{\min} n}} + \frac{\sqrt{L_F R \cdot \pi_{\min} n}}{\pi_{\min} n} = 2\sqrt{\frac{L_F R}{\pi_{\min} n}}.$$

The floor $\lfloor K^\star \rfloor$ introduces an additive $O(1/(\pi_{\min} n))$ term (the gap between $K^\star$ and $\lfloor K^\star \rfloor$ shifts the variance term by at most $1/(\pi_{\min} n)$), which is dominated by the $O(n^{-1/2})$ rate. □

A.12 Proof of Theorem 8 (T5.2 matching lower bound)

Proof. We construct a distribution achieving both lower bounds simultaneously.

Bias lower bound. Consider the family $F_{k,\sigma}(s) = \Phi(s - L_F \sigma)$, where Φ is the standard normal CDF. This satisfies 4 with Lipschitz constant L_F: for any σ_1, σ_2,

$$\sup_s |F_{k,\sigma_1}(s) - F_{k,\sigma_2}(s)| = \sup_s |\Phi(s - L_F\sigma_1) - \Phi(s - L_F\sigma_2)| \leq L_F |\sigma_1 - \sigma_2|,$$

where the inequality uses $\|\Phi'\|_\infty = \phi(0) = 1/\sqrt{2\pi} < 1$.

In any bin of width $\Delta_b = R/K$, the mixture CDF F_{kb} averages over shifted CDFs. Because the shift is monotone in σ, the CDF at the lower boundary $F_{k,e_{b-1}}$ and upper boundary F_{k,e_b} straddle the mixture:

$$\sup_s |F_{kb}(s) - F_{k,e_{b-1}}(s)| \;\geq\; \frac{L_F \Delta_b}{2} = \frac{L_F R}{2K}.$$

(The factor $1/2$ comes from the average shift relative to one endpoint being at least half the bin width.)

Variance lower bound. For any cell with n_{kb} calibration points, the conformal quantile's rank-based coverage is exactly $\lceil (n_{kb}+1)(1-\alpha) \rceil / (n_{kb}+1)$, which lies in $[1-\alpha,\, 1-\alpha+1/(n_{kb}+1)]$. When F_{kb} is continuous and the test point's rank is uniformly distributed, the gap $1/(n_{kb}+1)$ is attainable (up to ceiling effects). For the worst cell, $n_{kb} \leq \lceil \pi_{\min} n/K \rceil$, so the gap is at least $1/(2\lceil \pi_{\min} n/K \rceil + 1)$.

Combined lower bound. For any K:

$$G(K) \;\geq\; \max\left(\frac{L_F R}{2K},\; \frac{1}{2\lceil \pi_{\min} n/K \rceil + 1} \right).$$

Optimizing over K: the first term is $\Theta(1/K)$ and the second is $\Theta(K/n)$. They balance at $K = \Theta(\sqrt{n})$, giving

$$\min_K G(K) = \Omega\left(\sqrt{\frac{L_F R}{\pi_{\min} n}} \right),$$

matching the $O(n^{-1/2})$ upper bound of T5.1 up to constants. □

Interpretation. The $O(n^{-1/2})$ rate is dimension-free because T5 conditions on the 1-D summary $\hat{\sigma}(x)$, not on the full feature vector. By contrast, RLCP [Hore and Barber, 2025] achieves local coverage at rate $O(n^{-2/(d+2)})$, which is impractical at $d = 7{,}731$. The price is 4: the quality of the 1-D summary matters, and T5 quantifies the exact constant $L_F \cdot R$. In the homoscedastic limit $L_F = 0$, the oracle $K^\star = 0$ and no binning is needed — T1/T2 suffice.

B Data and training details

B.1 TraitGym benchmark

We use TraitGym [Benegas et al., 2025], a standardized evaluation suite for variant effect prediction. Two matched datasets:

- **Mendelian (matched_9)**: 338 ClinVar pathogenic variants + 3,042 matched benign controls = 3,380 total across 113 OMIM phenotypes. Matching on consequence type, MAF, and TSS distance (9 strata).
- **Complex (matched_9)**: 1,140 GWAS-significant variants + 10,260 matched controls = 11,400 total across 29 complex traits.

Chrom-LOO protocol. Test chromosomes: $\{17, 18, 19, 20, 21, 22, X\}$ (TraitGym default). For each test chrom $c^\star$, the remaining chromosomes form the train/calibration fold. Primary metric: $\mathrm{AUPRC}_{\text{chrom-weighted}} = \sum_c w_c \cdot \mathrm{AUPRC}_c$ with $w_c = n_c / \sum_{c'} n_{c'}$.

B.2 Input features

Table 5: Features used in HCCP aggregator. All pre-computed as per TraitGym conventions.

Feature	Source	Dim
CADD v1.7	Kircher et al. [2014]	1
Borzoi_L2	Linder et al. [2025] zero-shot L2-norm	3,846
Borzoi_L2_L2	Linder et al. [2025] L2-of-L2	3,846
GPN-MSA_LLR	Benegas et al. [2023] log-likelihood ratio	1
GPN-MSA_absLLR	Benegas et al. [2023] absolute LLR	1
GPN-MSA_InnerProducts	Benegas et al. [2023] inner products	36
CADD+GPN-MSA+Borzoi (full)		**7,731**
CADD+Borzoi		**7,693**

B.3 Aggregator hyperparameters

Base classifier $\hat{p}$. `HistGradientBoostingClassifier` (scikit-learn 1.4): `max_depth=2`, `max_iter=100`, `learning_rate=0.1`, `class_weight=balanced`, `random_state=42`, `early_stopping=False`. Feature imputation: mean via `SimpleImputer`. No feature scaling (GBM is scale-invariant).

Heteroscedastic head $\hat{\sigma}$. `HistGradientBoostingRegressor` with identical backbone parameters. Target: $r_i = |Y_i - \hat{p}^{(-C_i)}(X_i)|$ (chrom-LOO OOF residuals). Loss: squared error (Gaussian NLL approximation). Floor: $\hat{\sigma}(x) = \max(\hat{\sigma}_{\text{raw}}(x), \epsilon)$ with $\epsilon = 10^{-4}$.

Conformal calibration. Score: $s(x, y) = |y - \hat{p}(x)|/(\hat{\sigma}(x) + \epsilon)$. Bin count: $K = \hat{K}_{\text{CV}}$ (adaptive, §5.3); fallback $K = 5$ for fixed-K baselines. Cell minimum $n_{k,b} \geq 5$; fallback to class-pooled $\hat{q}_k$ otherwise. Miscoverage: $\alpha = 0.10$.

DEGU-lite ensemble. $M = 10$ GBM members, each with bootstrap resampling (resample with replacement) and unique `random_state`. Ensemble uncertainty: $\hat{\sigma}_{\text{DEGU}}(x) = \mathrm{std}_{m=1}^{M} \hat{p}_m(x)$. Same backbone as aggregator.

B.4 Within-cell KS audit

To assess A2-cell (score stationarity within Mondrian cells across chromosomes), we ran 155 two-sample KS tests (5 bins $\times$ 2 classes $\times$ up to 22 calibration chroms, restricted to cells with $n \geq 20$). Results: 12.3% rejection at $p < 0.05$ (expected 5% under null); max KS statistic 0.24, mean 0.06. Conclusion: A2-cell is approximately but not exactly satisfied; T3′ (robust form) absorbs the residual via the chrom-TV term.

B.5 Compute

All experiments on a single machine with 8$\times$ NVIDIA T4 GPUs (SM 7.5, no BF16). The HCCP pipeline (aggregator + $\hat{\sigma}$ head + conformal calibration) runs entirely on CPU via scikit-learn; a full chrom-LOO sweep takes <4 CPU-hours per feature set. DEGU-lite ensemble ($M = 10$) takes ≈35 CPU-minutes (Mendelian) and ≈75 CPU-minutes (Complex). Figure and proof generation: negligible.

C Per-partition tables

C.1 Per-chromosome coverage

Table 6 reports per-chromosome coverage for the HCCP Mondrian predictor (CADD+GPN-MSA+Borzoi, $\alpha = 0.10$, $K = 5$).

Table 6: Per-chromosome coverage under Mondrian-($y \times \hat{\sigma}$-bin), Mendelian dataset. Target: 0.90. Chromosomes 17–22, X are test chroms under chrom-LOO.

Chrom	n	Coverage	|gap|
1	210	0.857	0.043
2	230	0.896	0.004
3	155	0.945	0.045
5	100	0.850	0.050
6	150	0.800	0.100
7	210	0.943	0.043
8	140	0.943	0.043
9	240	0.842	0.058
10	190	0.879	0.021
11	160	0.944	0.044
12	150	0.933	0.033
13	210	0.848	0.052
14	100	0.900	0.000
16	80	0.913	0.013
17	150	0.900	0.000
19	200	0.828	0.073
20	50	0.900	0.000
22	20	0.950	0.050
X	250	0.958	0.058
Overall	**3380**	**0.902**	**0.002**

C.2 K-sweep ablation

Table 7 validates Theorem 7 (T5.1) by sweeping $K \in \{2, 3, 5, 8, 10, 15, 20, 30\}$.

C.3 DEGU-lite vs HCCP head-to-head

Table 8 compares HCCP's supervised $\hat{\sigma}$ against DEGU-lite's ensemble $\hat{\sigma}$ under identical Mondrian-($y \times \hat{\sigma}$-bin) conformal calibration and feature sets.

Interpretation. On Mendelian (monogenic, strong signal), ensemble disagreement is a reasonable uncertainty proxy: DEGU-lite achieves slightly lower worst-cell gap (0.033 vs. 0.041). On Complex

Table 7: K-sweep on Mendelian and Complex (CADD+GPN-MSA+Borzoi). Worst-cell gap = $\max_{k,b} |\mathrm{cov}_{k,b} - 0.90|$. $\hat{K}_{\mathrm{CV}} = 3$ (Mendelian), $\hat{K}_{\mathrm{CV}} = 2$ (Complex).

	Mendelian		Complex	
K	Worst gap	Mean gap	Worst gap	Mean gap
2	0.233	0.059	**0.006**	**0.002**
3	**0.047**	**0.012**	0.010	0.004
5	0.285	0.041	0.042	0.007
8	0.627	0.067	0.058	0.010
10	0.700	0.059	0.208	0.017
15	0.733	0.049	0.600	0.034
20	0.733	0.047	0.600	0.030
30	0.344	0.031	0.567	0.034

Table 8: HCCP vs DEGU-lite: same features (CADD+GPN-MSA+Borzoi), same Mondrian partition, different $\hat{\sigma}$ source. Worst gap = max per-$\hat{\sigma}$-bin deviation from target 0.90.

$\hat{\sigma}$ source	Dataset	Cov	$\mathrm{Cov}\|_{Y=1}$	Worst gap	Mean gap
HCCP (learned)	Mendelian	0.902	0.905	0.041	0.022
DEGU (ensemble)	Mendelian	0.901	0.917	**0.033**	**0.020**
HCCP (learned)	Complex	0.901	0.908	**0.013**	**0.007**
DEGU (ensemble)	Complex	0.901	0.907	0.057	0.028

(polygenic, dominant aleatoric noise), supervised residual regression provides $4.3\times$ better worst-cell coverage (0.013 vs. 0.057). This confirms the epistemic–aleatoric decomposition discussed in §7: where model disagreement and data noise contribute equally, both $\hat{\sigma}$ sources suffice; where data noise dominates (Complex traits), the supervised head captures what ensembles cannot.

C.4 Cross-dataset transfer

Table 9: Cross-dataset transfer (CADD+Borzoi). M→C: train on Mendelian, test on Complex. C→M: reverse. T4 proxy bound = $1 - \alpha - d_{\mathrm{TV}}$.

Direction	AUPRC	Coverage	Gap	TV proxy	T4 bound
M→C	0.187	0.738	-0.162	0.158	0.742
C→M	0.654	0.901	$+0.001$	0.158	0.742

Directional asymmetry. C→M transfer preserves coverage (0.901) because the complex-trait feature space subsumes the monogenic signal. M→C collapses (0.738) because monogenic variants occupy a narrow subspace. The Barber T4 proxy bound (0.742) is tight for M→C (observed 0.738, within 0.004) and conservative for C→M.